Supporting Information for

An Anionic Metal−Organic Framework as Ultrafast Single-ion Conductor for Exceptional Performance Rechargeable Zinc Battery

Qingchun Xia,[1,3,*] Xuxiao Ma,[1] Pengtao Qiu,[1] Guozan Yuan,[2,*] and Xuenian Chen[1,3,4,*]

[1]*Henan Key Laboratory of Boron Chemistry and Advanced Energy Materials, Key Laboratory of Green Chemical Media and Reactions, Ministry of Education, School of Chemistry and Chemical Engineering, Henan Normal University, Xinxiang, Henan 453007, China.*

[2]*School of Chemistry and Chemical Engineering, Anhui University of Technology, Ma'anshan, Anhui 243032, China.*

[3]*State Key Laboratory of Coordination Chemistry, School of Chemistry and Chemical Engineering, Nanjing University, Nanjing 210093, China.*

[4]*College of Chemistry, Zhengzhou University, Zhengzhou, Henan 450001, China.*

*e-mail: xiaqingchun@htu.edu.cn; guozan@ahut.edu.cn; xuenian_chen@zzu.edu.cn

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1 Materials and general procedures

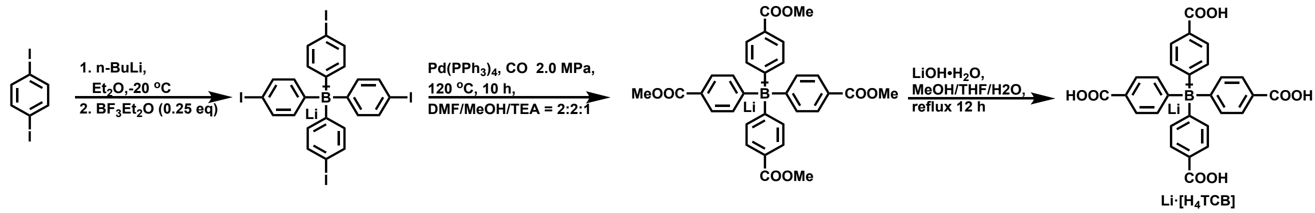
All reagents and solvents used in these studies are commercially available and used without further purification.

Single-crystal XRD data for **1** were collected on a Bruker SMART Apex II CCD-based X-ray diffractometer with Cu-Kα radiation (λ = 1.54178 Å) at 150 K. The empirical absorption correction was applied by using the SADABS program (G. M. Sheldrick, SADABS, program for empirical absorption correction of area detector data; University of Göttingen, Göttingen, Germany, 1996). The structure was solved by direct methods with SHELXS-2018 and refined with SHELXL-2018 using OLEX 1.2. In the structure, all the non-hydrogen atoms were refined by full-matrix least-squares techniques with anisotropic displacement parameters, and the hydrogen atoms were geometrically fixed at the calculated positions attached to their parent atoms, treated as riding atoms. Contributions to scattering due to these highly disordered solvent molecules were removed using the *SQUEEZE* routine of *PLATON*; structures were then refined again using the data generated.

The structure was then refined again using the data generated. Crystal data and details of the data collection are given in **Table S1**. CCDC 2222634 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from the Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data request/cif

2 Experimental procedure for synthesis and analysis

2.1 Synthesis of the ligand Li·[H_4L]



2.1.1 Synthesis of lithium tetrakis(4-iodophenyl)borate

1,4-Diiodobenzene (32.9 g, 100 mmol) was dissolved in Et_2O (150 mL). The solution was cooled to −20 ºC and treated dropwise with a solution of n-BuLi (40 mL, 2.5 M in hexane, 100 mmol) while maintaining the temperature below −20 ºC. The resulting mixture was kept at −10 ºC for 30 min, and BF_3·**Et_2O** (2.3 ml, 18.7 mmol) was then added. After that, the cooling bath was removed and the mixture was allowed to warm to room temperature and left overnight. The mixture was filtered to get a solid, and the solid was washed with ether several times, which was then redissolved into the acetone (100 mL). Filter and remove the insoluble LiF, the filtrate was concentrated under reduced pressure to remove the acetone, leaving the desired lithium tetrakis(4-iodophenyl)borate 14.9 g with 72% yield. ^{1}H NMR (600 MHz, DMSO-d_6): δ 6.93-6.90 (m, 8H), 7.31 (d, 8H, J = 7.8 Hz). ^{11}B NMR (193 MHz, DMSO-d_6): δ −7.31. ^{13}C NMR (151 MHz, DMSO-d_6): δ 88.3, 134.5, 137.9, 161.7 (m).

2.1.2 Synthesis of lithium tetrakis (4-(methoxycarbonyl) phenyl) borate

To a mixture of DMF (52 mL), MeOH (52 mL) and triethylamine (26 mL) in an autoclave, lithium tetrakis (4-iodophenyl) borate (13.0 g, 15.37 mmol) and Pd(PPh_3)$_4$ (3.3 g, 3.0 mmol, 20 mol%) were added. Seal the autoclave and press the CO to 2 MPa, the autoclave was stirred and heated at 120 ºC for 10 hours. After cooling to room temperature, the solvents were removed under reduce pressure, and the residue was redissolved into ethyl acetate. The mixture was then filtered to get the filtrate, and the filtrate was concentrated under vacuum and then purified by column chromatography on silica gel (EtOAc/ Actione, 10/1, v/v) to afford the desired product lithium tetrakis (4-(methoxycarbonyl) phenyl) borate 6.43 g with 75% yield. ^1H NMR (600 MHz, DMSO-d_6): δ = 7.62 (d, J = 7.8 Hz, 8H), 7.27 (m, 8H), 3.78 (s, 12H). ^{11}B NMR (193 MHz, DMSO-d_6): δ = −6.37. ^{13}C NMR (151 MHz, DMSO-d_6): 169.83 (m), 167.39, 135.24, 126.84, 124.13, 51.49.

2.1.3 Synthesis of lithium tetrakis(4-carboxyphenyl)borate (Li·[H_4L])

To a mixture of MeOH (40 mL), THF (40 mL) and water (40 mL) in a round-bottom flask, lithium tetrakis (4-(methoxycarbonyl) phenyl) borate (4.1 g, 7.14 mmol), LiOH·H_2O (1.43g, 34.08 mmol) were added. The mixture was refluxed for 16 h. MeOH and THF were removed by evaporation, and the residue was diluted with H_2O and washed with EA for 3 times. The aqueous phase was acidified with concentrated HCl, and the precipitate was collected by filtration and then dried under vacuum to afford 2.32 g of brown solid Li·[H_4**L**] with 65% yield. ^1H NMR (600 MHz, DMSO-d_6): δ = 7.59 (d, J = 8.1 Hz, 8H), 7.25 (m, 8H). ^{11}B NMR (193 MHz, DMSO-d_6): δ = −6.43. ^{13}C NMR (151 MHz, DMSO-d_6):169.37 (m), 168.68, 135.19, 127.09, 125.21.

2.2 Synthesis of 1

A mixture of $ZrOCl_2 \cdot 8H_2O$ (161 mg, 0.5 mmol), Li·[H_4**L**] (251 mg, 0.5 mmol), DMF (60 mL), TFA (10 mL) was sealed in a 100 mL vial with a screw cap and heated at 100 ^oC for 24 h. After that, the colorless crystals were collected, washed with MeOH, and dried under vacuum at 80 ^oC to afford the **1** (305 mg, 68%). The product can be best formulated as $Li_2 \cdot [Zr_6O_4(OH)_8(H_2O)_4\mathbf{L}_2] \cdot nG$ based on single-crystal analysis, and the letter G in the formula represents the disordered guest molecules that occupy in the frameworks, which may include H_2O, DMF, and/or TFA. Elemental analysis for **1**: Anal (%). Calcd for $C_{56}H_{56}B_2Zr_6O_{32}Li_2$; C, 36.88; H, 3.09. Found: C, 39.02; H, 3.34. ICP measurement indicated the molar ratio of Li:B:Zr is about 0.84:1:2.89. IR (KBr, cm^{-1}): 1645 (w), 1580 (w), 1502 (w), 1401(s), 1279 (w), 1242(w), 1178 (m), 1034 (m), 761 (m), 711 (w), 640 (s), 454 (s).

2.3 Synthesis of $\mathbf{1^{Zn}}$

To a Schlenk tube, 60 mg of **1** and 2 mL 2.0 M aqueous solution of $Zn(NO_3)_2$ were added. The Schlenk tube was then immersed in the oil bath and incubated at 80 ^oC for 4 h. Subsequently, the crystals were collected via vacuum filtration, and the aforementioned ion exchange experiment was repeated every 4 hours for a total of 5 cycles until no free Li^+ was detected by ICP-MS. The final product was collected, washed with water, and dried under vacuum at 120 ^oC to afford $\mathbf{1^{Zn}}$ as a white free-flowing powder. Elemental analysis for $\mathbf{1^{Zn}}$: Anal (%). Calcd for $C_{56}H_{56}B_2Zr_6O_{32}Zn$; C, 35.87; H, 3.01. Found: C, 36.38; H, 3.73. ICP measurement indicated the molar ratio of Zn:B:Zr is about 0.47:1:3.11. IR (KBr, cm^{-1}): 1652(m), 1588(w), 1502(m), 1402(s), 1108(w), 1014(w), 1022 (w), 821 (w), 763 (m), 720 (w), 642 (m), 455 (s).

2.4 Weighing analysis of $\mathbf{1^{Zn}}$

$\mathbf{1^{Zn}}$ was first subjected to Soxhlet extraction using methanol as the eluent for comprehensive purification. After that, $\mathbf{1^{Zn}}$ was then degassed under vacuum at 160 °C for 10 h to ensure the complete removal of residual solvent molecules within the framework. The activated sample was then weighed to determine its dry mass (m_0).

The activated $\mathbf{1^{Zn}}$ was immersed in DI water for 24 h under ambient conditions. The hydrated sample was subsequently filtered and dried in an oven at 65°C for 12 hours to afford a free-flowing powder, which was then weighed to determine its mass (m_1). The water content in the pore was calculated via mass differential analysis ($\Delta m = m_1 - m_0$), and the water percentage content was then expressed as: $H_2O\% = (m_1 - m_0)/m_0 \times 100$.

The free-flowing powder was then placed at room temperature and weighed again after one week (t_1) and one month (t_2) to calculate the water loss within the pore. The water loss was expressed as: $Water\ loss\ (\%) = [(m_1 - m_t)/(m_1 - m_0)] \times 100$.

2.5 Chemical stability

The activated **1**, 100 mg for each batch, was immersed in about 50 mL aqueous solution of 3 M HCl, 1 M NaOH, and 1 mM K_3PO_4 at room temperature for 24 h, respectively. The treated samples were then filtered, washed with DI water (3 times), acetone (3 times), and dried under vacuum at 100 °C for 10 h before PXRD, gravimetric analysis, ^{11}B NMR, and N_2 sorption measurements. The recoveries after the vacuum drying were 98.7, 101.7, and 102.4 mg, respectively.

2.6 Synthesis of $Zn[Me_4L]_2$

To a flask, lithium tetrakis (4-(methoxycarbonyl) phenyl) borate (1.12 g, 2.0 mmol) was suspended in 50 mL water. Zinc nitrate hexahydrate (12.0 g, 40 mmol) was added to this suspension, and then stirred at room temperature overnight. The exchanged $Zn[Me_4L]_2$ was collected by filtration, and the above ion-exchange operation was repeated four times to ensure the complete exchange of Li ions. The final exchanged $Zn[Me_4L]_2$ was collected by filtration and dried under vacuum.

3 Electrochemical measurements

3.1 Cyclic voltammetry (CV) measurement of 1

Under a force of 5 MPa, 50 mg **1** was compressed into a pellet with a diameter ranging from 1.0 to 1.4 mm. Subsequently, the pellet was assembled into an airtight Swagelok-type cell to form an asymmetric SS|$\mathbf{1^{Zn}}$|Zn cell which was then employed to perform the cyclic voltammetry (CV) measurement at a scan rate of 5 mV/s with a voltage range from –0.5 to 5.0 V (vs. Zn/Zn^{2+}) at room temperature.

3.2 Linear sweep voltammograms (LSV) measurement of 1

The linear sweep voltammetry (LSV) measurement was conducted with the aforementioned asymmetric SS|$\mathbf{1^{Zn}}$|Zn cell under a sweep rate of 0.5 mV/s in a voltage range from –0.3 V to 6.0 V (vs. Zn/Zn^{2+}) at room temperature.

3.3 Ionic conductivity of $\mathbf{1^{Zn}}$

50 mg $\mathbf{1^{Zn}}$ was ground into a fine powder and then pressed into a thin pellet with a thickness of 1.0 to 1.4 mm under a force of 5 MPa. The final thickness of the pellet was measured with a vernier caliper, and the pellet was then sealed in an airtight Swagelok-type cell between two stainless steel blocking electrodes to form a symmetric SS|$\mathbf{1^{Zn}}$|SS cell for ionic conductivity measurement.

The alternating current (AC) impedance analysis was performed using a two-probe method with an Autolab AUT88031 potentiostat/galvanostat over the frequency range of 10^{-2} to 10^6 Hz, with an input voltage amplitude of 100 mV. Data was recorded every 10 °C between –40 to 120 °C. Each temperature was held for 1 h to reach thermal equilibrium. At each temperature point, the ionic conductivities (σ)

were determined according to the following equation:

$$\sigma = \frac{L}{AR} \qquad (1)$$

where L and A represent the thickness (cm) and the area (cm^2) of the pellet, respectively. And R, which was extracted directly from the impedance plots, is the bulk resistance of the sample (Ω). The activation energy (E_a) was calculated with the Nernst-Einstein relation:

$$\sigma = \frac{\sigma_0}{T} \exp\left(-\frac{E_a}{kT}\right) \qquad (2)$$

where σ_0 is a pre-exponential factor, T is the temperature, E_a is the activation energy, and k is the Boltzmann constant.

3.4 Electrical conductivity of $\mathbf{1^{Zn}}$

Direct current (DC) polarization measurements were carried out to determine the electronic conductivity. 50 mg $\mathbf{1^{Zn}}$ was ground into a fine powder and then pressed into a thin pellet with a thickness of 1.0 to 1.4 mm under a force of 5 MPa. The final thickness of the pellet was measured with a vernier caliper, and the pellet was then sealed in an airtight Swagelok-type cell between two stainless steel blocking electrodes to form a symmetric SS|$\mathbf{1^{Zn}}$|SS cell for electronic conductivity measurement. The cells were connected to the Autolab AUT88031 potentiostat/galvanostat in combination with a low-current option. Measurements were recorded at a constant voltage of 2 V, and the currents were recorded for 2000 s The electronic conductivity was calculated to be 1.54×10^{-9} S/cm.

3.5 Ionic transference number of $\mathbf{1^{Zn}}$

The ionic transference number was evaluated using a potentiostatic polarization method at room temperature. The $\mathbf{1^{Zn}}$ pellet assembled into an airtight Swagelok-type cell between two Zn foils to form a symmetric Zn|$\mathbf{1^{Zn}}$|Zn cell. The direct current (DC) polarization measurement was performed on this cell using an Autolab AUT88031 potentiostat/galvanostat with an applied polarized voltage ranging from 10 mV to 50 mV. The AC impedance of the symmetric cell was conducted over the frequency range 1 MHz–1 Hz, with an input voltage amplitude of 100 mV. The ionic transference number t was determined according to the following equation:

$$t = \frac{I^S(\Delta V - I^0 R^0)}{I^0(\Delta V - I^S R^S)} \qquad (3)$$

where I^S is the steady-state current, I^0 is the initial current, ΔV is the applied potential, R^0 and R^S are the interfacial resistances before and after the polarization, respectively.

3.6 The Ohm's law method to determine the electrical conductivity

The electrical conductivity, σ, is generally obtained by fitting the linear region of a current−voltage (I−V) curve to Ohm's law. From the Ohm's law:

$$R = \frac{V}{I}$$

For the $\mathbf{1^{Zn}}$,

$$R = \frac{L}{\sigma A}$$

$$A = \pi r^2$$

L and r are the thickness and radius of the pellet, respectively.

Therefore,

$$\sigma = \frac{L}{\pi r^2} \times \frac{I}{V} \qquad (4)$$

A direct current (DC) voltage ranging from −0.8 to 0.8 V was then applied to the symmetric SS/$\mathbf{1}^{\mathbf{Zn}}$/SS cell (Figure S19), and their corresponding currents were measured to draw the I−V curve. The slope of the fitted lines was then calculated to determine the electrical conductivity σ using the above formula, and the value was evaluated to be 1.52×10^{-9} S/cm.

3.7 Galvanostatic cycling measurements

Galvanostatic cycling measurement was performed on the asymmetric Zn|$\mathbf{1}^{\mathbf{Zn}}$|Cu cell assembled in the CR2025 button cell. Galvanostatic cycling for this asymmetrical cell was performed using a multichannel battery testing system (LAND CT3001A) by sequentially charge and discharge.

The galvanostatic cycling measurement of the symmetric Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn cell was similar, and this symmetric Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn cell was also assembled in the CR2025 button cell.

3.8 Galvanostatic charge/discharge measurements of the quasi-solid ZIB

The $\mathbf{1}^{\mathbf{Zn}}$ electrolyte was prepared by making a slurry containing $\mathbf{1}^{\mathbf{Zn}}$: poly(vinylidene difluoride) (PVDF) in the weight ratio of 60:40 in 1-methyl-2-pyrrolidinone (NMP). The slurry was then cast onto stainless steel foil, followed by drying at 85 °C in a vacuum oven for 12 h.

The $NH_4V_4O_{10}$ cathode was prepared from a slurry of $NH_4V_4O_{10}$, Super-p, and PVDF in the weight ratio of 70:20:10 in NMP. The slurry was ground and then carefully coated on the carbon paper. The coated carbon paper was dried at 85 ℃ in a vacuum oven for 12 h, and pressed for 2 min under a pressure of 30 MPa. The weight loading of $NH_4V_4O_{10}$ was about 1~2 mg/cm^2.

The polished Zn foil, $\mathbf{1}^{\mathbf{Zn}}$ electrolyte, and $NH_4V_4O_{10}$ cathode were assembled into a CR2025 coin cell. The galvanostatic charge/discharge test of the quasi-solid ZIB was recorded by cycling between 0.3 to 1.6 V vs Zn^{2+}/Zn with a current density of 1 C, after a 6 h rest at 30 °C. The coin cell was measured by a LAND CT3001A instrument.

The MnO_2, PTCDA, and $Cu_3(HHTP)_2$ cathodes were prepared similarly.

3.9 ICP-MS analysis the composition of $\mathbf{1}^{\mathbf{Zn}}$ electrolyte

After the rate performances, we disassembled the Zn|$\mathbf{1}^{\mathbf{Zn}}$|Cu, Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn, and Zn|$\mathbf{1}^{\mathbf{Zn}}$|$NH_4V_4O_{10}$. The $\mathbf{1}^{\mathbf{Zn}}$ electrolytes were digested in a small beaker with 10 mL aqua regia, diluted by 10 wt% HNO_3, and transferred to a volumetric flask, washed the beaker three times, and then diluted with dilute nitric acid to 100 mL for ICP measurement. ICP-MS measurements indicated the molar ratios of the Zn:Zr after the rate performance were 0.146:1, 0.161:1, and 0.157:1, respectively, almost consistent with the original ratio of 0.151:1 for the pristine of $\mathbf{1}^{\mathbf{Zn}}$.

Table S1. Crystal data and structure refinement for 1.

Identification code	**1**
Empirical formula	$C_{28}H_{16}BO_{16}Zr_3$
Formula weight	892.88
Temperature (K)	170
Wavelength (Å)	CuKα (λ = 1.54184)
Crystal system	tetragonal
Space group	I4/m
Unit cell dimensions	a = 15.3170(3) Å b = 15.3170(3) Å c = 27.7377(6) Å $\alpha = \beta = \gamma = 90$
Volume (Å^3), Z	6507.6(3), 4
Density (calculated) (mg/m^3)	0.911
Absorption coefficient (mm^{-1})	4.200
F(000)	1748.0
θ range for data collection (º)	8.164 to 142.898
Limiting indices	$-10 \leq h \leq 18$ $-10 \leq k \leq 18$ $-33 \leq l \leq 16$
Reflections collected	7267
Independent reflections	3165 [R_{int} = 0.0334, R_{sigma} = 0.0471]
Completeness to theta	98%
Data / restraints / parameters	3165/0/114
Goodness-of-fit on F^2	1.072
Final R indices [I>2sigma(I)]	R_1 = 0.0468, wR_2 = 0.1379
R indices (all data)	R_1 = 0.0579, wR_2 = 0.1464
Largest diff. peak and hole (e.Å^{-3})	1.28/-1.18

4 Characterization

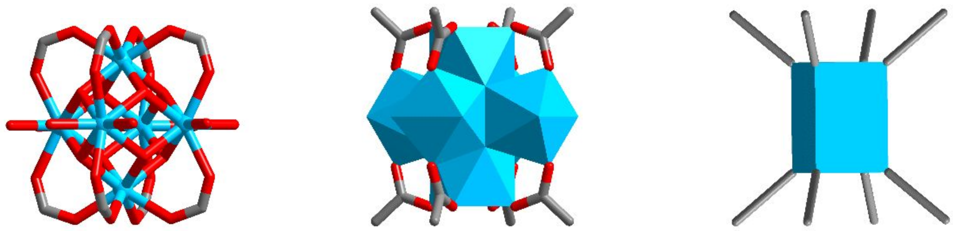


Figure S1. The asymmetric unit in **1**.

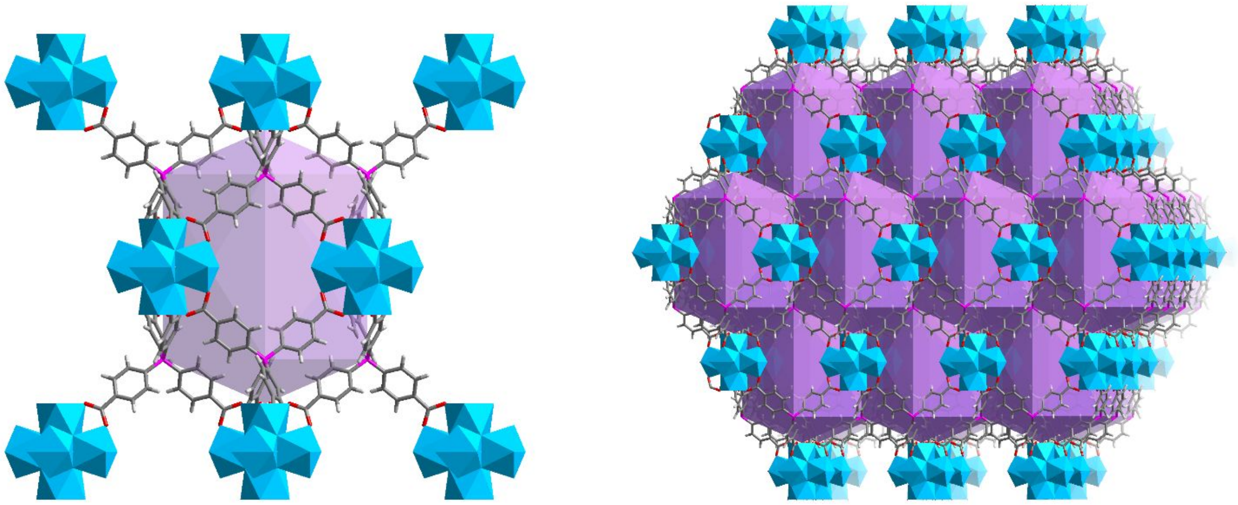


Figure S2. The coordination environments of the Zr_6 cluster.

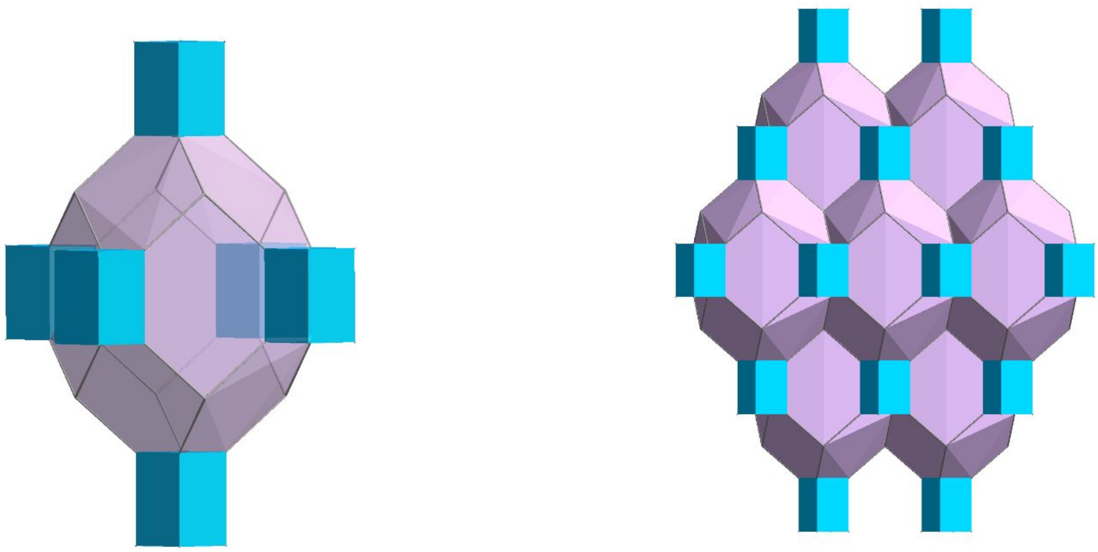


Figure S3. The packing mode and simplified dodecahedral cage in **1**.

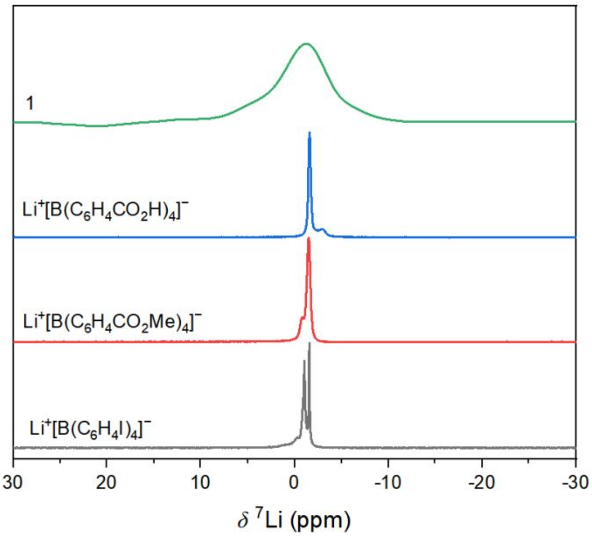


Figure S4. ^{7}Li MAS NMR spectra.

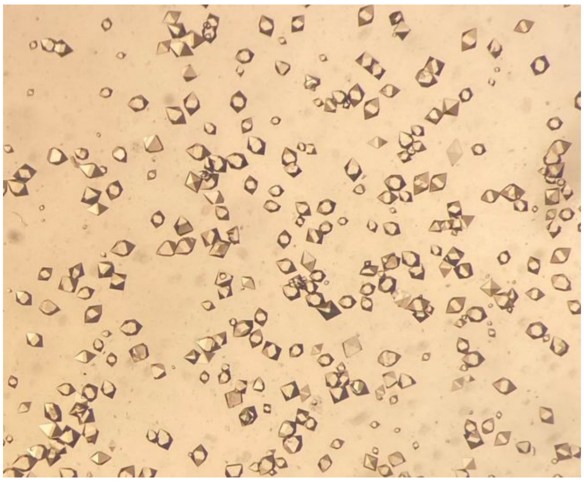


Figure S5. The micrographs of **1**.

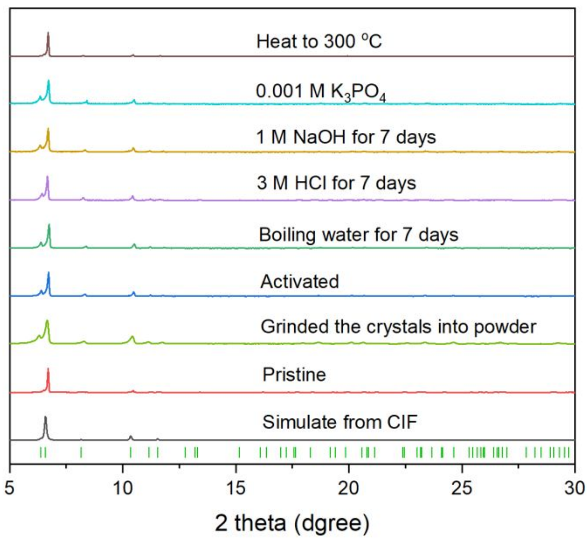


Figure S6. PXRD patterns of **1**.

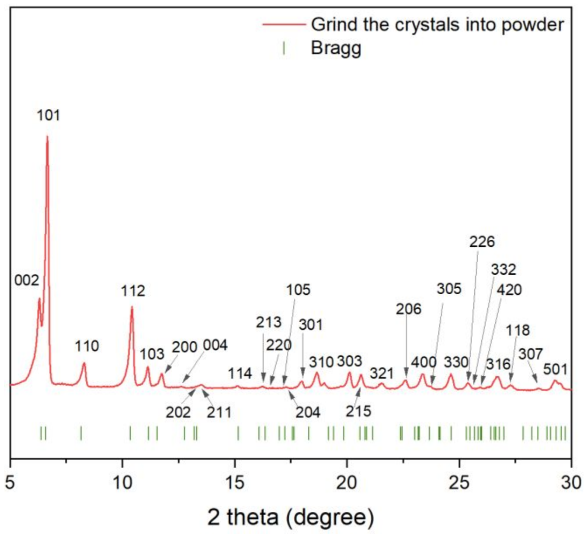


Figure S7. PXRD patterns of **1** after grinding into powder.

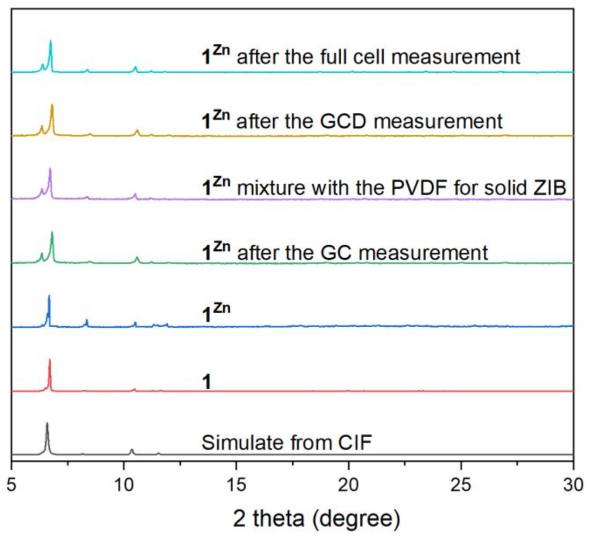


Figure S8. PXRD patterns of $\mathbf{1^{Zn}}$.

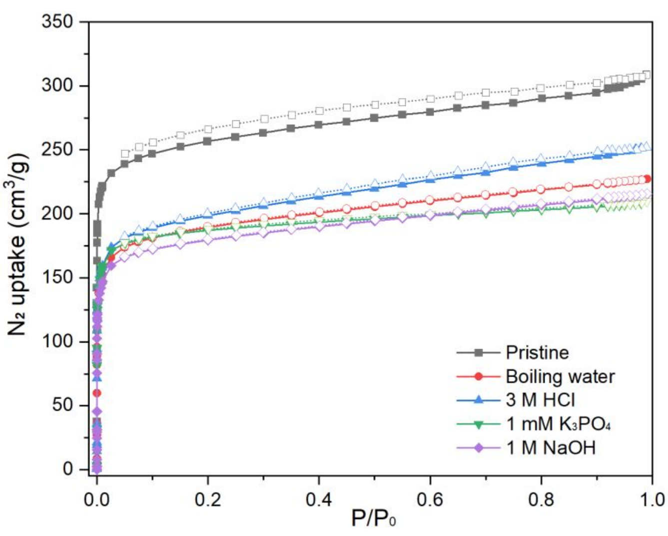


Figure S9. N_2 adsorption isotherms of **1** after various treatments.

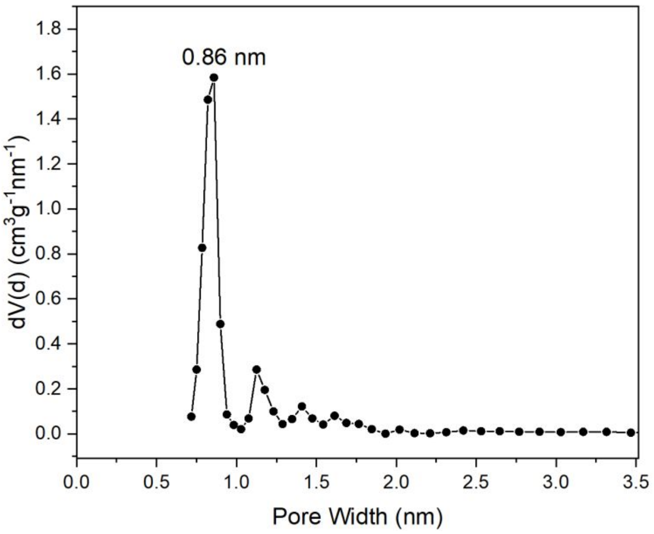


Figure S10. Pore size distribution profiles of **1**.

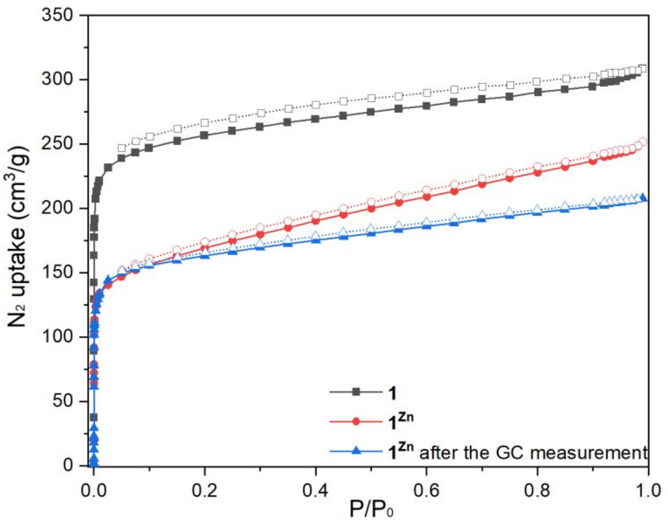


Figure S11. N_2 adsorption isotherms of $\mathbf{1^{Zn}}$.

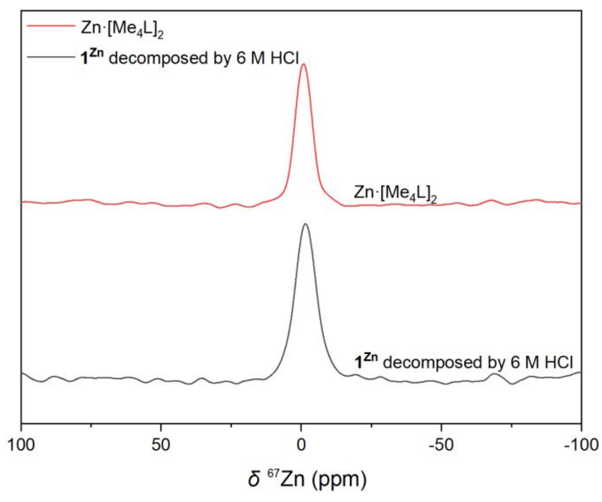


Figure S12. ^{67}Zn NMR spectra of $\mathbf{1^{Zn}}$ and Zn·[Me_4**L**]$_2$. ^{67}Zn NMR of $\mathbf{1^{Zn}}$ was conducted by decomposing $\mathbf{1^{Zn}}$ with 6 M HCl to dissolve in water.

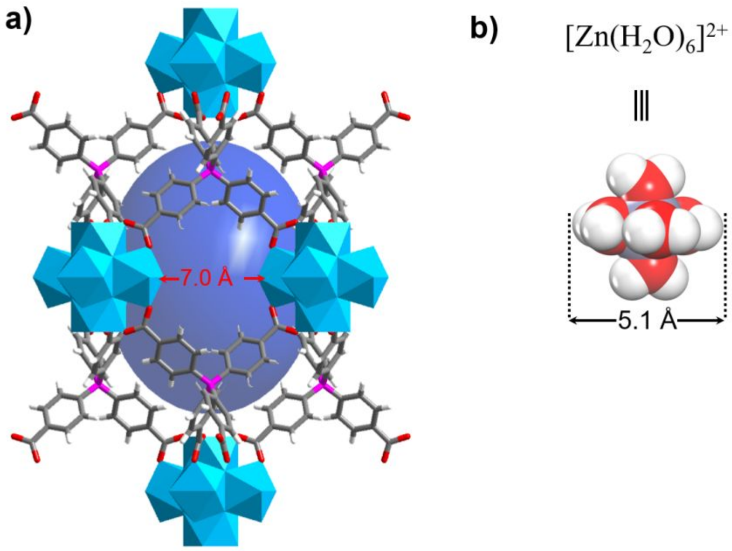


Figure S13. The size comparison among the pore of the framework and $Zn(H_2O)_6^{2+}$ ion. **a)** The pore width of the framework, and **b)** the size of the $Zn(H_2O)_6^{2+}$ion.

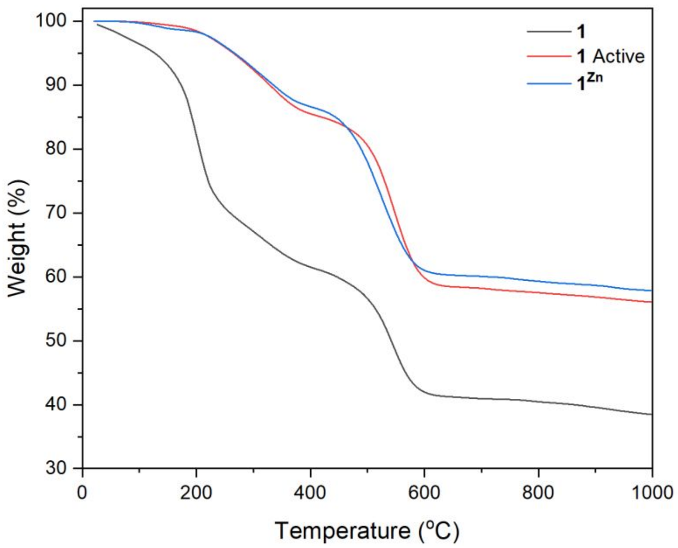


Figure S14. TGA curve.

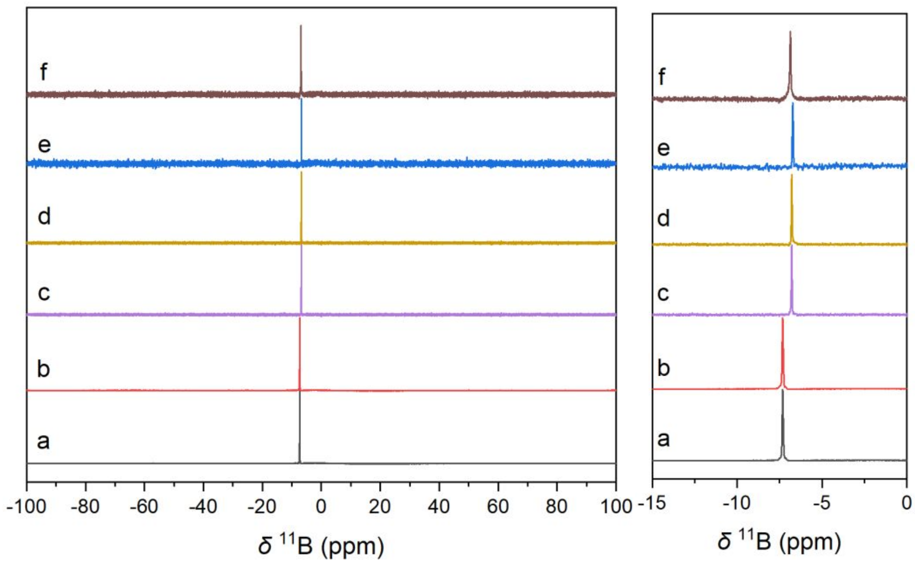


Figure S15. [1]B NMR of the decomposed **1** after the treatment. a, b) ^{11}B NMR of the ligand $Li^+[B(C_6H_4COOH)_4]^-$ and immersing in 6 M HCl for more than 24h, respectively. c, d, and e) ^{11}B NMR of **1** after immersing 3 M HCl, 1 M NaOH, and 1 mM K_3PO_4, for more than 24h, respectively. f) ^{11}B NMR of **1** after heating to 300 ^{o}C.

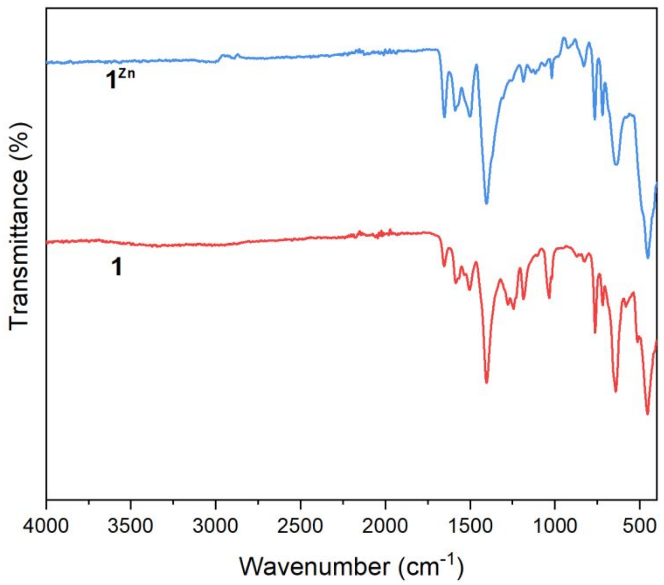


Figure S16. IR spectrum**.**

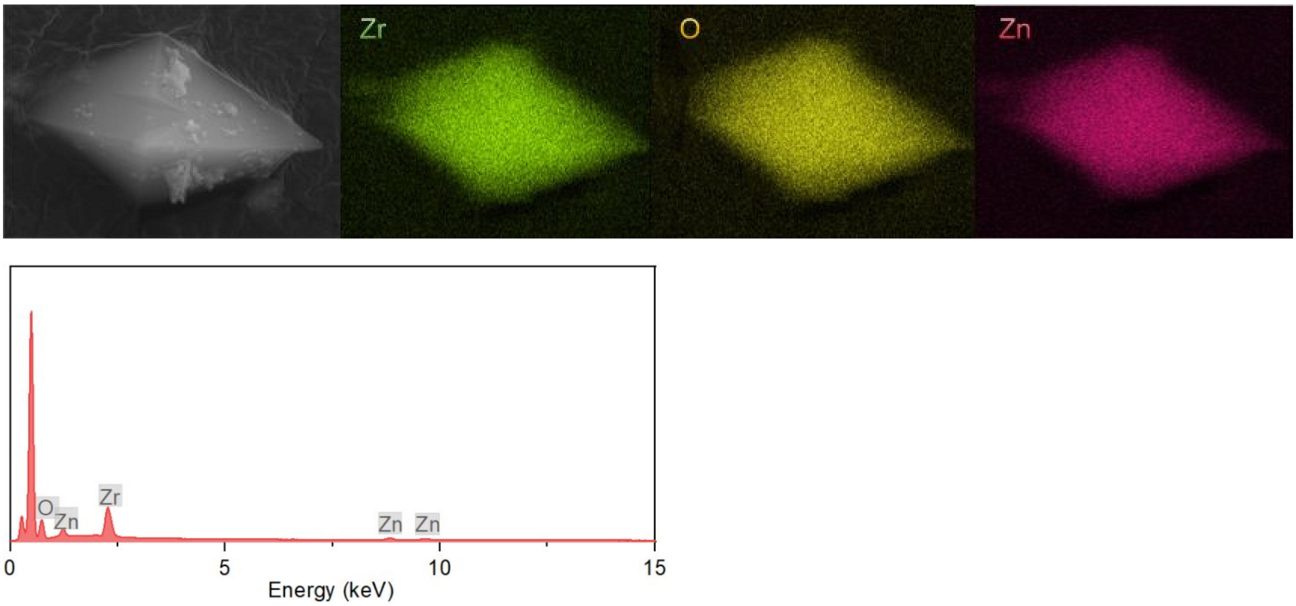


Figure S17. SEM and SEM-EDS mapping of $\mathbf{1}^{\mathbf{Zn}}$.

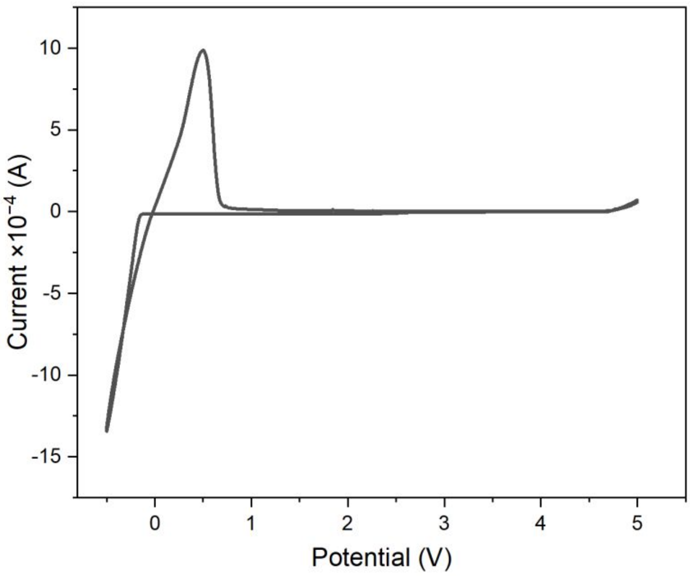


Figure S18. Cyclic voltammetry (CV) test for $\mathbf{1}^{\mathbf{Zn}}$. CV of $\mathbf{1}^{\mathbf{Zn}}$ displayed a wide window of electrochemical stability, no significant current flow corresponding to electrolyte decomposition was observed up to 5.0 V. Only cathodic and anodic currents corresponding to Zinc deposition ($Zn^{2+} + 2e^- \rightarrow Zn$) and stripping ($Zn \rightarrow Zn^{+} + 2e^-$) were observed near 0 V vs Zn.

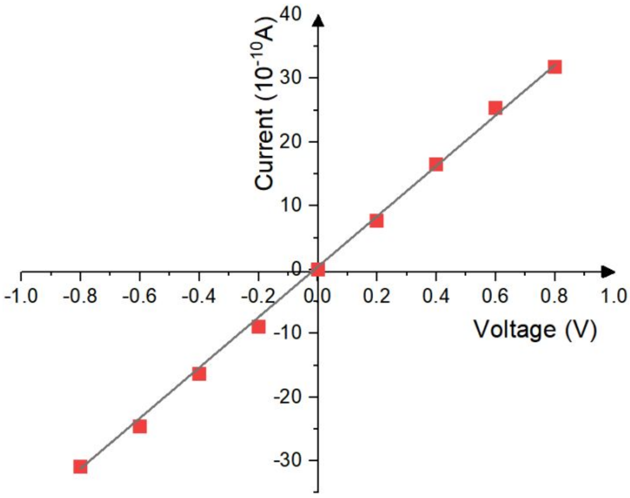


Figure S19. The electronic conductivity of 1^{Zn} determined by direct current (DC) polarization measurements.

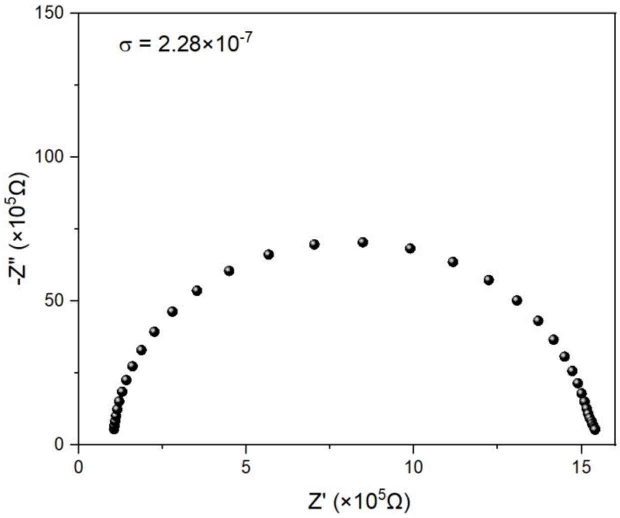


Figure S20. The ionic conductivity of MOF-841.

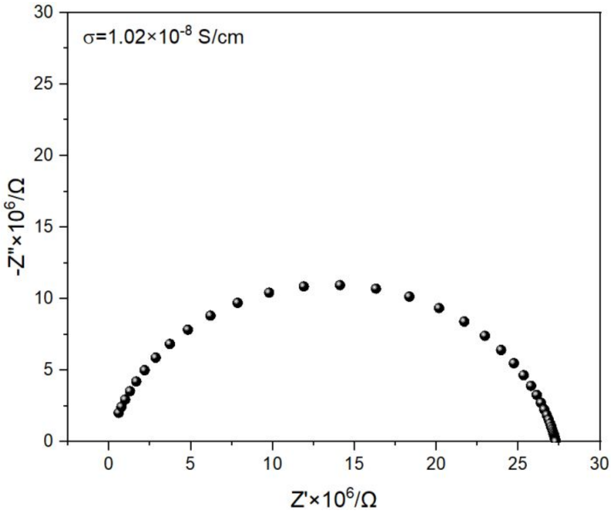


Figure S21. The ionic conductivity of Zn·[Me_4L]$_2$.

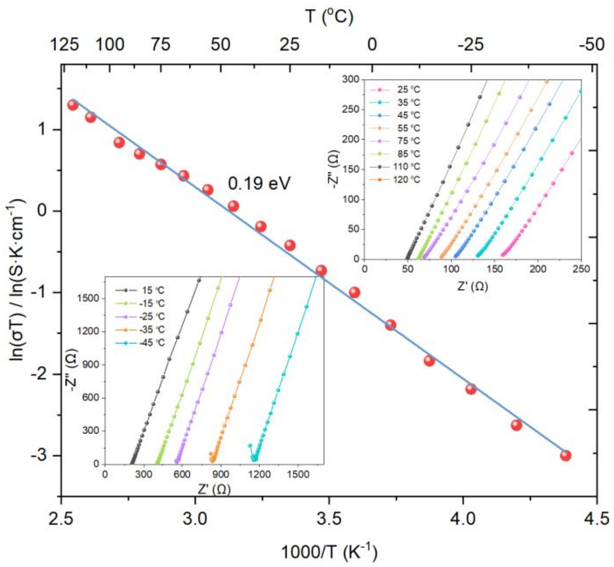


Figure S22. The activation energy of $\mathbf{1^{Zn}}$.

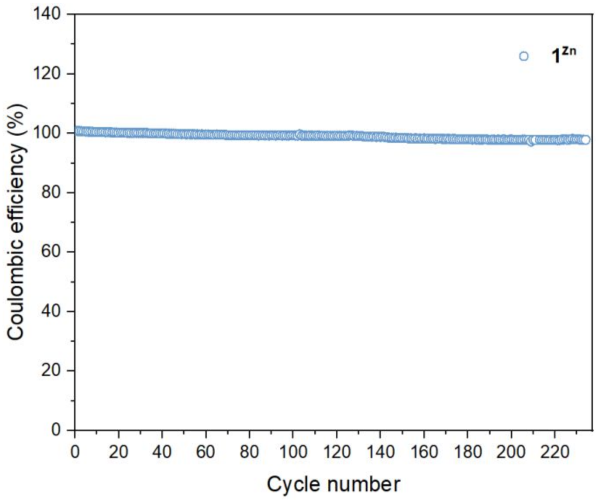


Figure S23. Rate performance of the asymmetric Zn|$\mathbf{1^{Zn}}$|Cu cell.

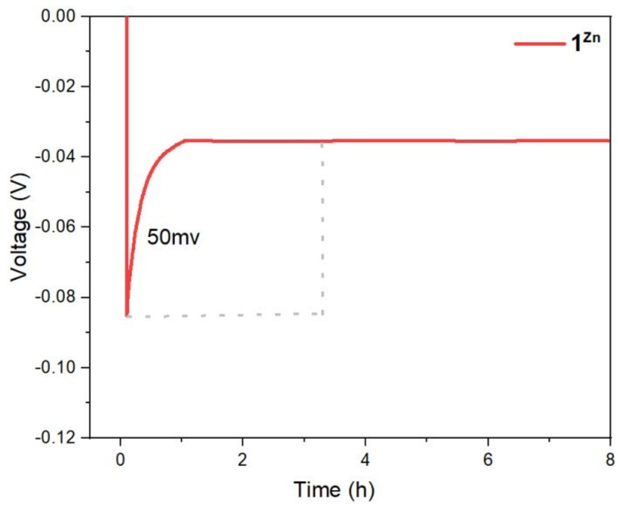


Figure S24. The nucleation overpotential of the asymmetric Zn|$\mathbf{1^{Zn}}$|Cu cell.

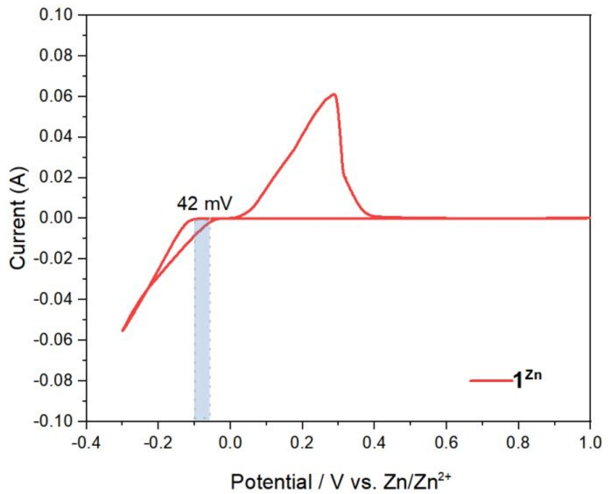


Figure S25. Cyclic voltammetry (CV) measurement of Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn cell.

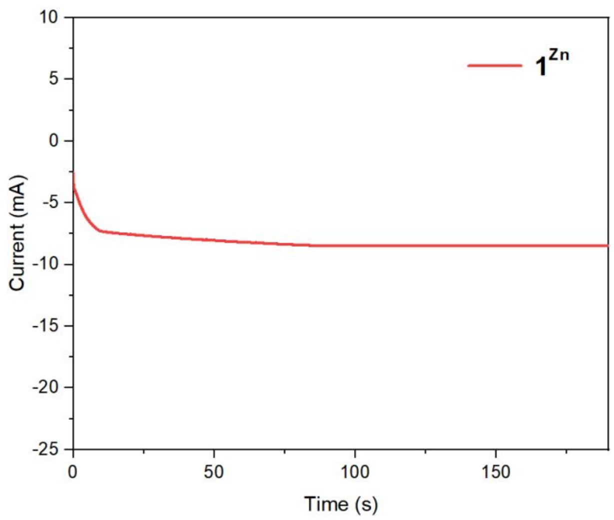


Figure S26. Curves of chronoamperometry for the symmetric Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn cell.

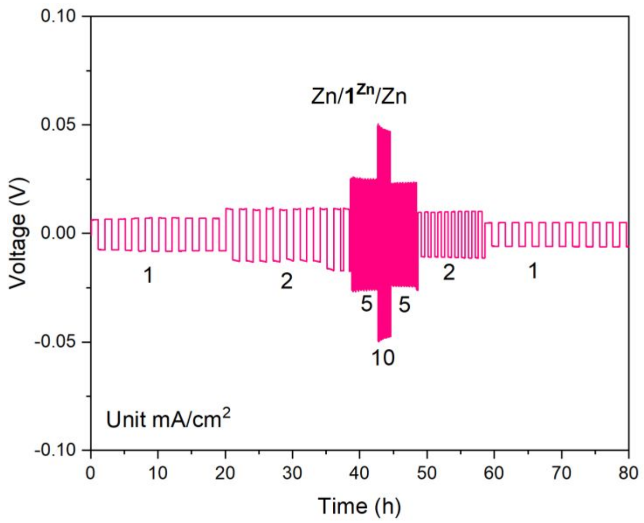


Figure S27. Rate performance of the symmetric Zn|$\mathbf{1}^{\mathbf{Zn}}$|Zn cell at current densities from 1 to 10 mA/cm^2.

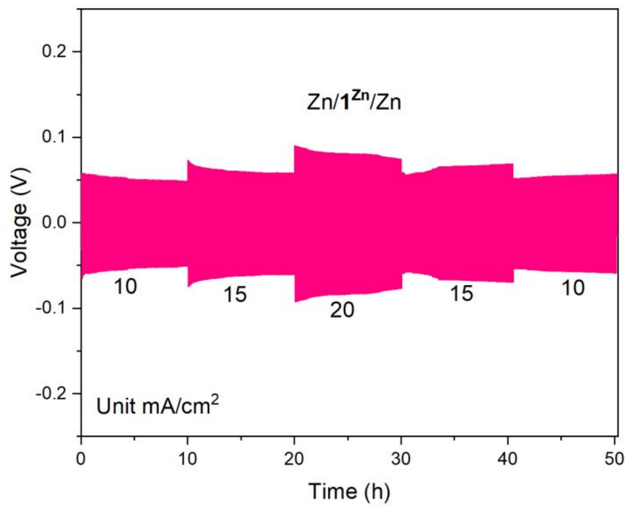


Figure S28. Rate performance of the symmetric Zn|**1**$^{\mathbf{Zn}}$|Zn cell at current densities from 10 to 20 mA/cm^2.

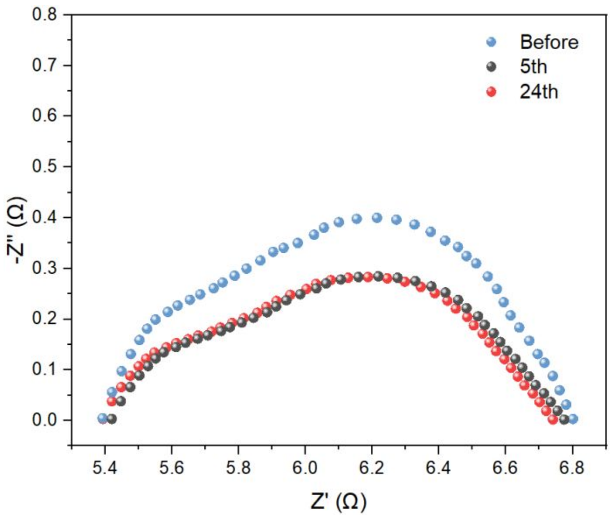


Figure S29. Impedance changes during the GC measurement.

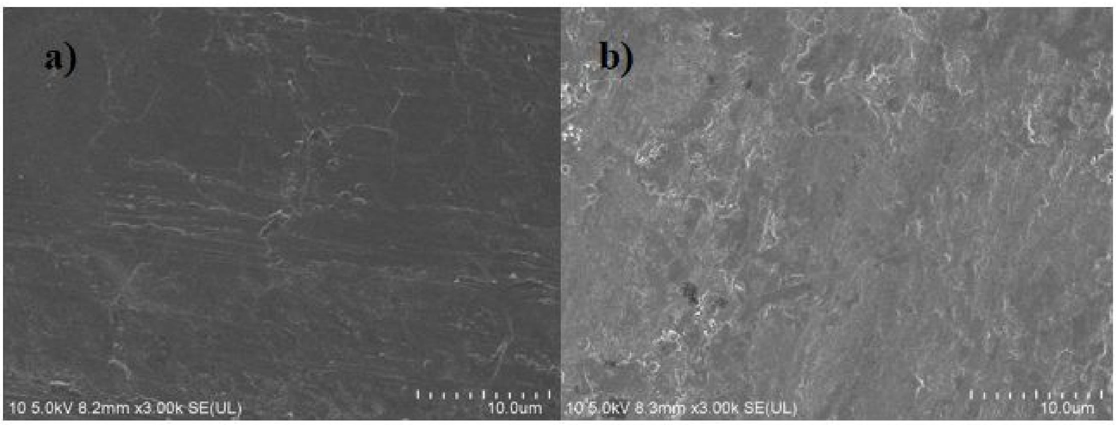


Figure S30. SEM morphology of the Zn metal surface before (**a**) and after (**b**) the GC measurement.

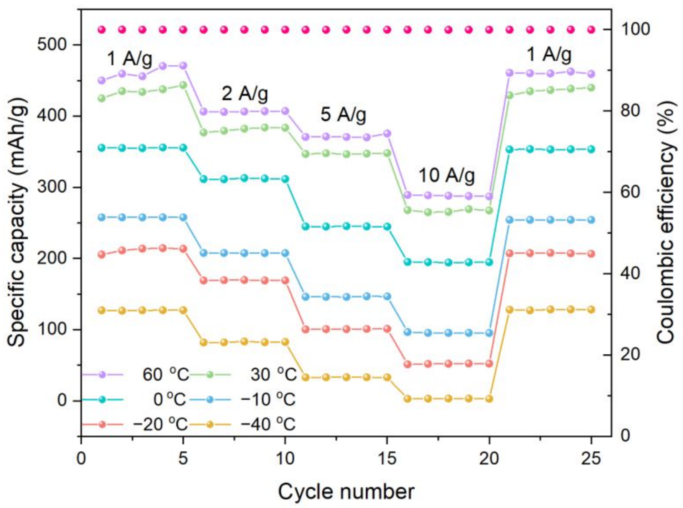


Figure S31. Rate and cycling performance of the Zn|**1**^{Zn}|$NH_4V_4O_{10}$ battery at the current density from 1 to 10 A/g at 60, 30, 0, −10, −20 and −40 °C, respectively.

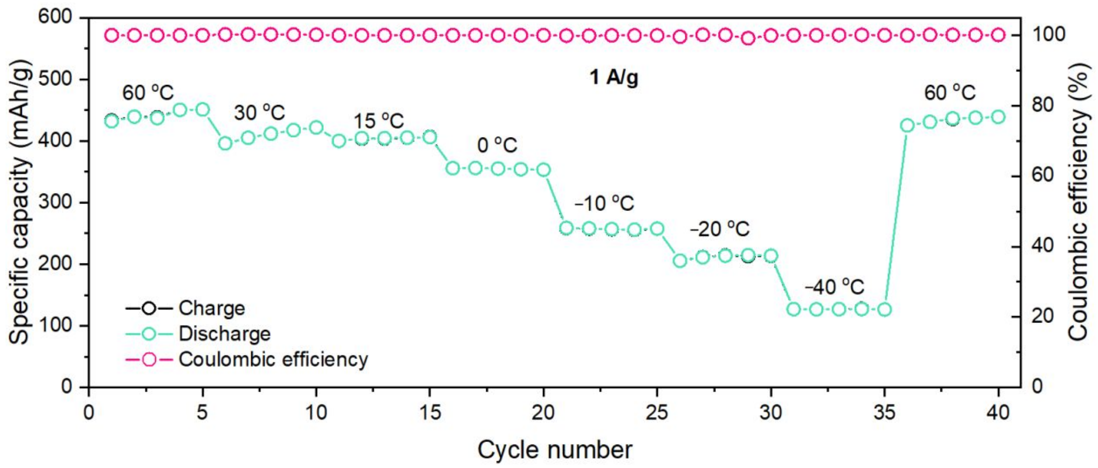


Figure S32. GCD performance of the Zn|**1**^{Zn}|$NH_4V_4O_{10}$ battery at 1 A/g stepwise decrease of temperature from 60 to −40 °C, and then stepwise increase to 60 °C.

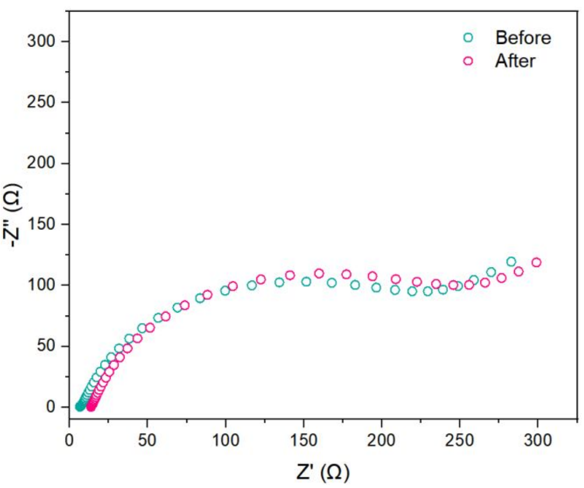


Figure S33. Impedance changes of the Zn|**1**^{Zn}|$NH_4V_4O_{10}$ during the GCD measurement.

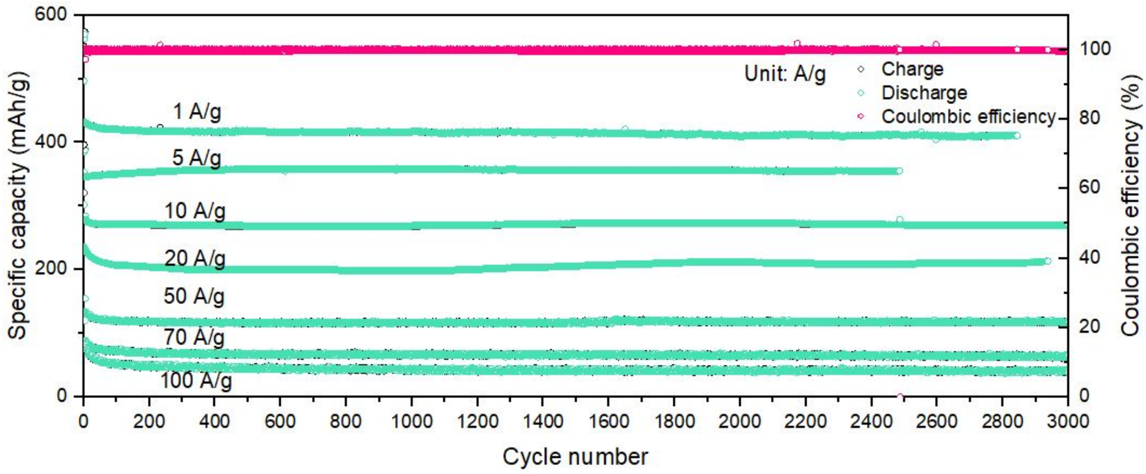


Figure S34. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at 1, 5, 10, 20, 50, 70, and 100 A/g at room temperature, respectively.

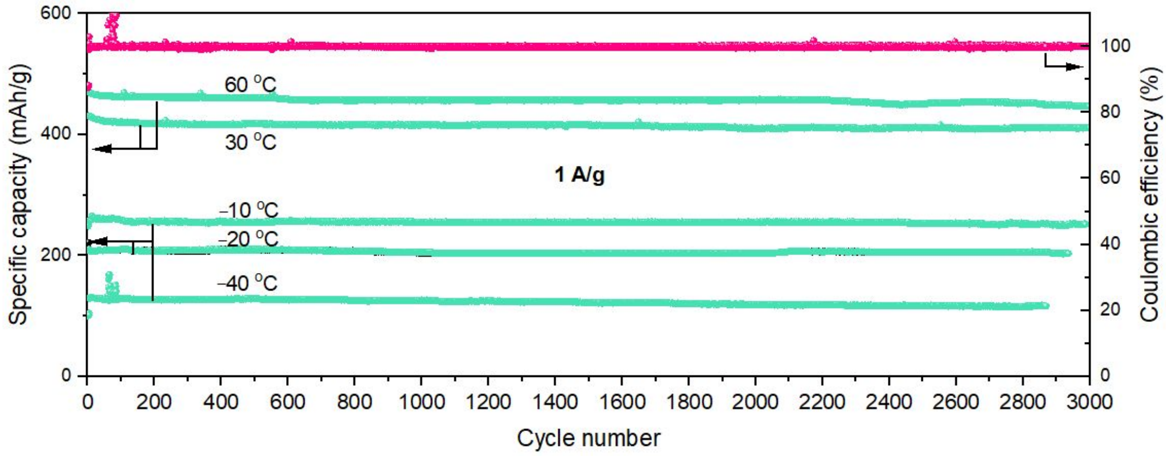


Figure S35. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at 1 A/g at −40, −20, −10, 30, and 60 °C.

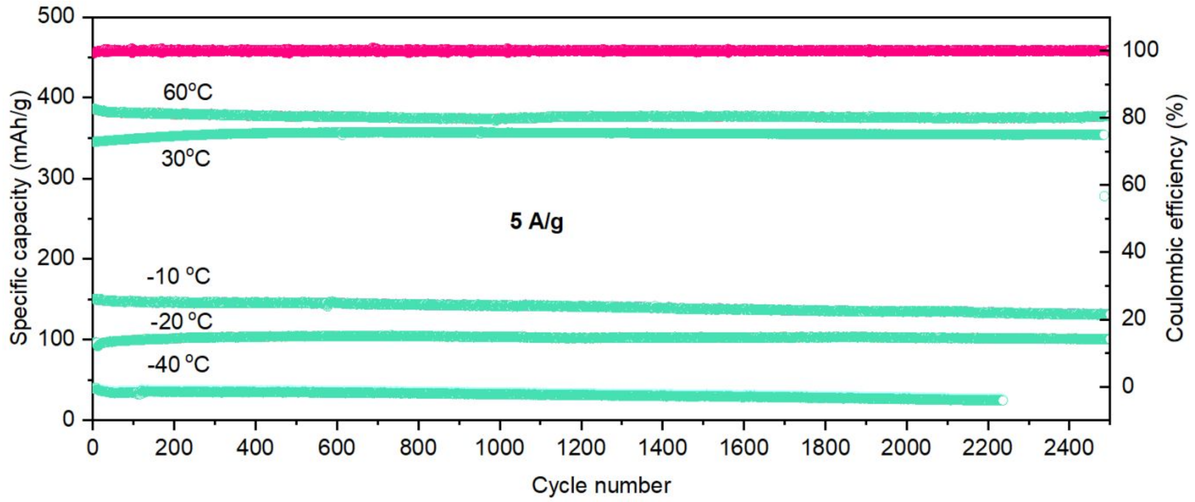


Figure S36. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at 5 A/g at −40, −20, −10, 30, and 60 °C.

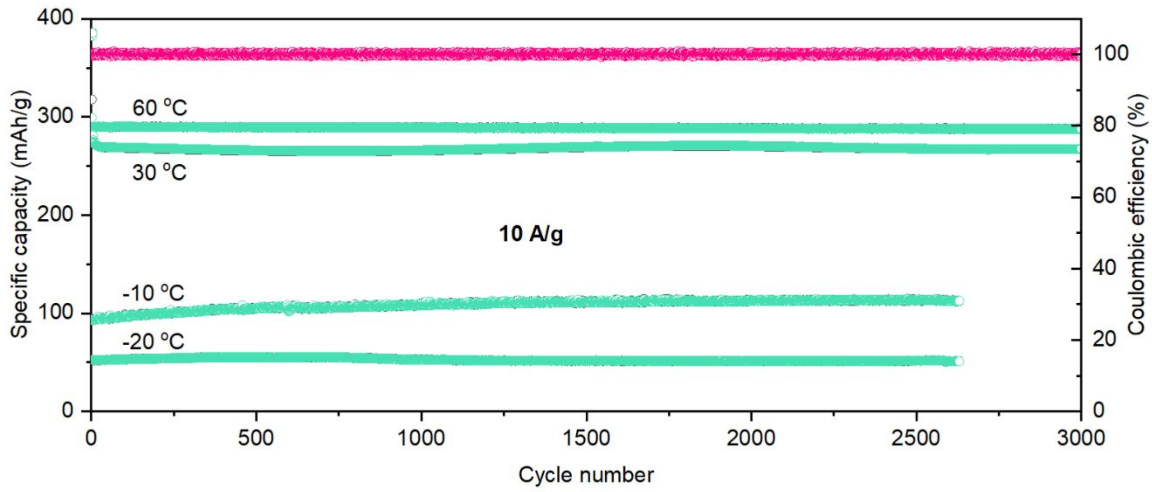


Figure S37. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at 10 A/g at −20, −10, 30, and 60 ^{o}C.

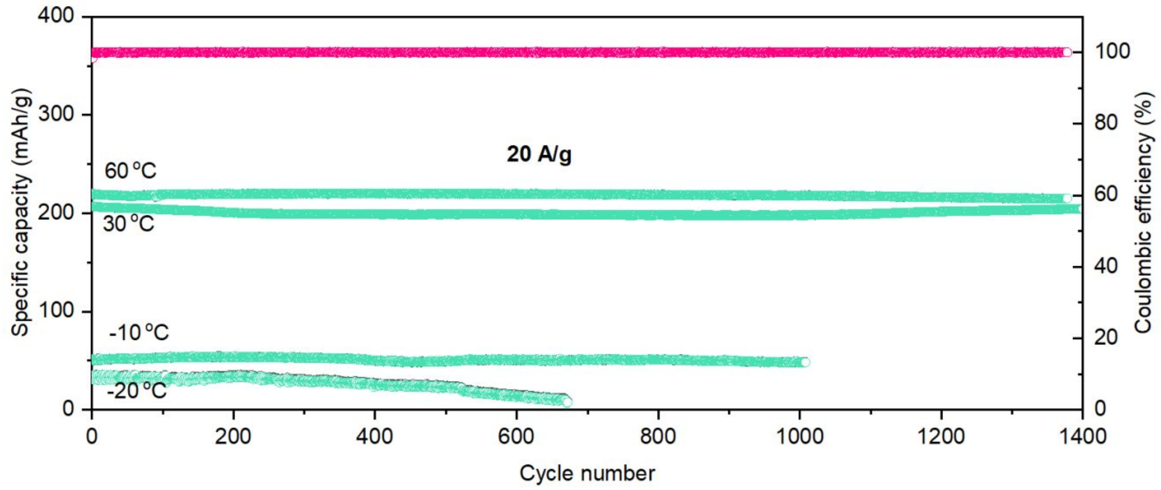


Figure S38. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at 20 A/g at −20, −10, 30, and 60 ^{o}C.

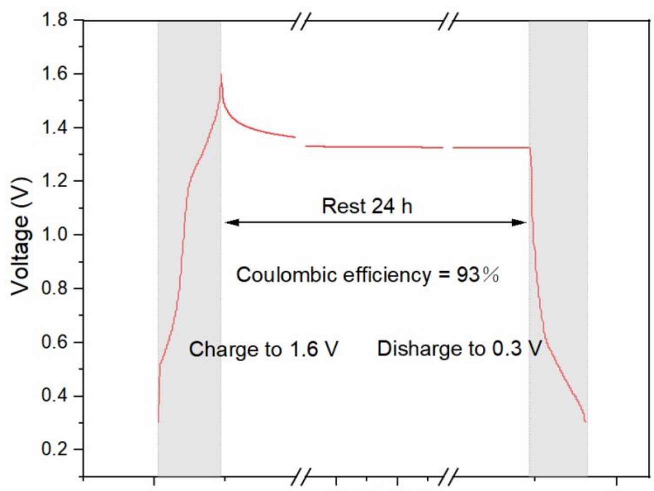


Figure S39. Self-discharge tests of the Zn|**1**$^{\mathbf{Zn}}$|$NH_4V_4O_{10}$ battery at room temperature.

Figure S40. Open circuit voltage of the pouch cell Zn|$\mathbf{1}^{\mathbf{Zn}}$|$NH_4V_4O_{10}$.

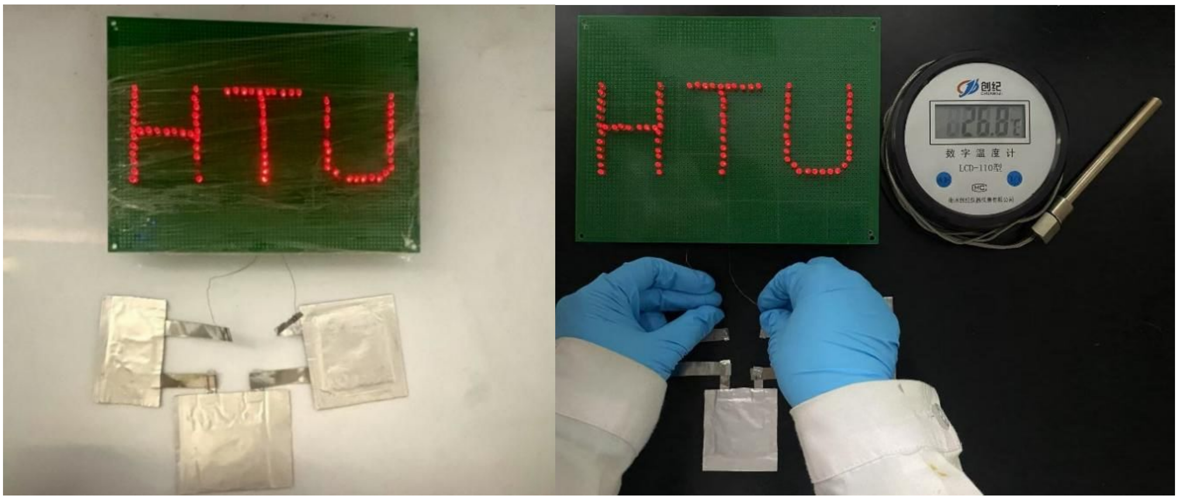


Figure S41. Photograph of the assembled pouch cells powering the array of LEDs at room temperature.

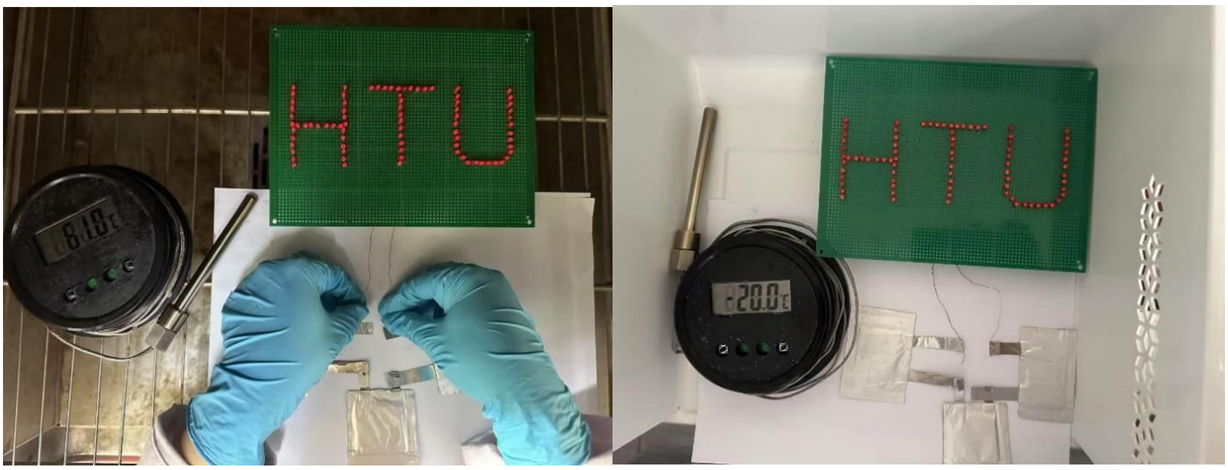


Figure S42. Photograph of the assembled pouch cells powering the array of LEDs at 60 ^{o}C and -20 ^{o}C.

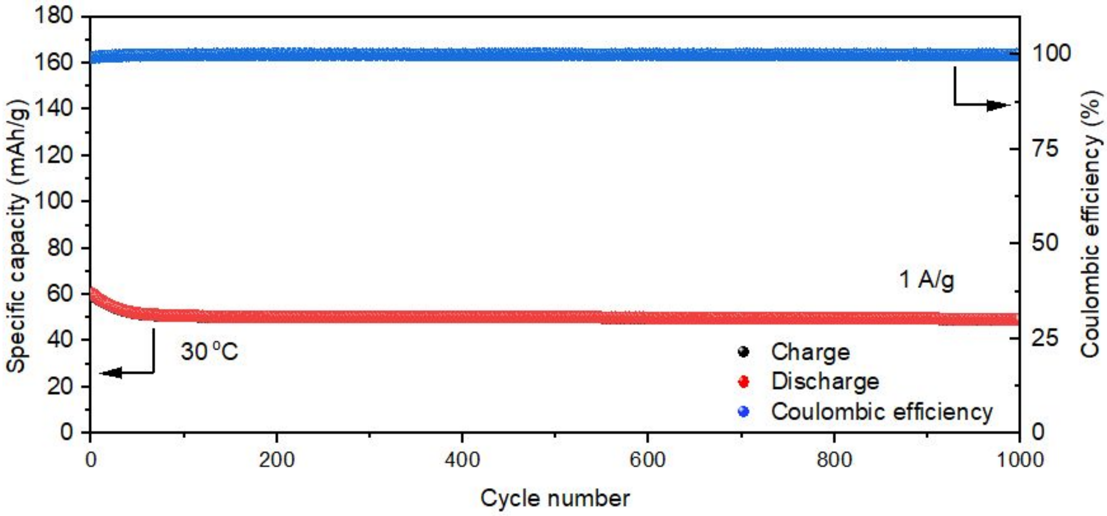


Figure S43. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|MnO_2 battery at 1 A/g at room temperature

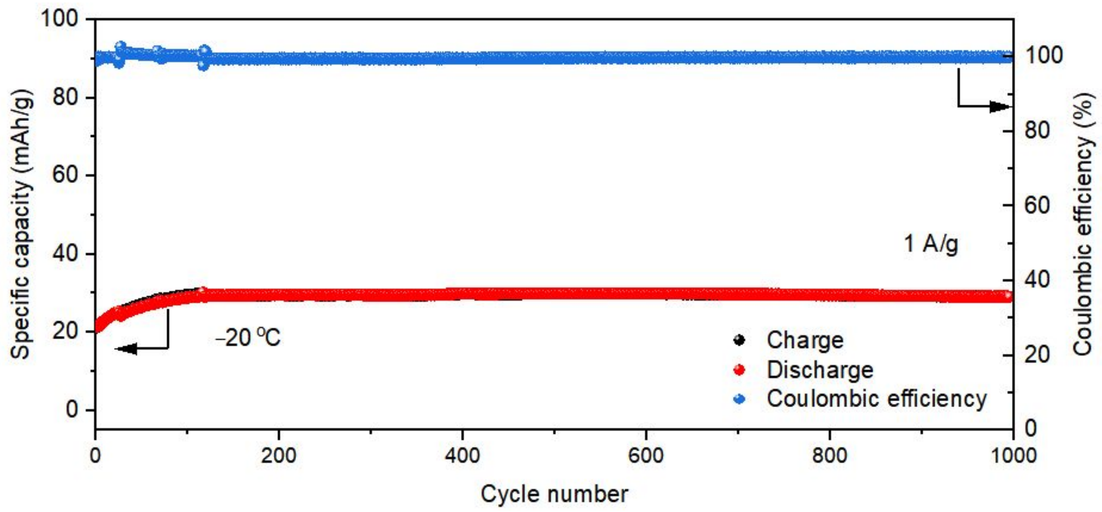


Figure S44. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|MnO_2 battery at 1 A/g at -20 ^{o}C

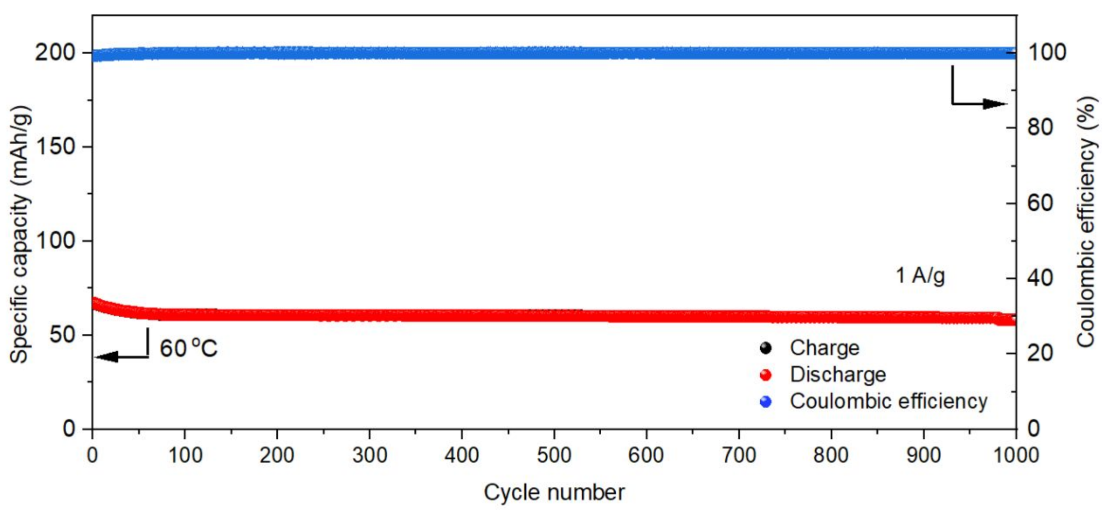


Figure S45. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$|MnO_2 battery at 1 A/g at 60 ^{o}C

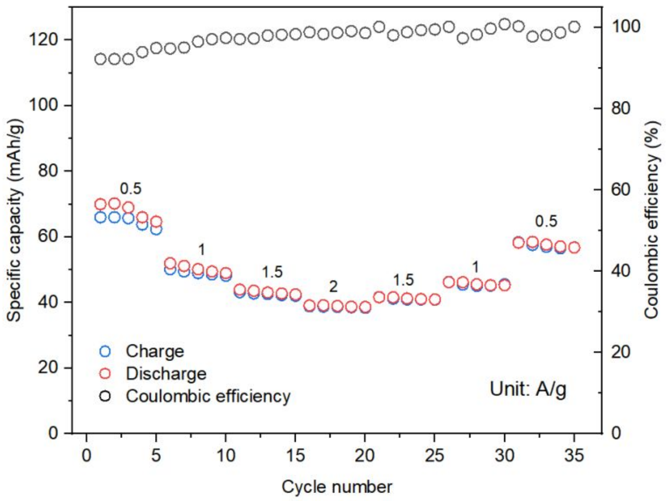


Figure S46. Galvanostatic charge-discharge curves of the Zn|**1**$^{\mathbf{Zn}}$| MnO_2 battery at different rates at room temperature.

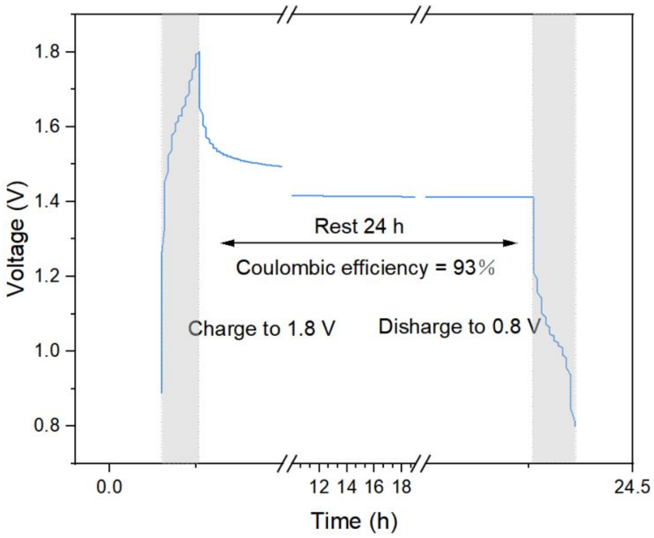


Figure S47. Self-discharge tests of the Zn|**1**$^{\mathbf{Zn}}$|MnO_2 battery at room temperature.

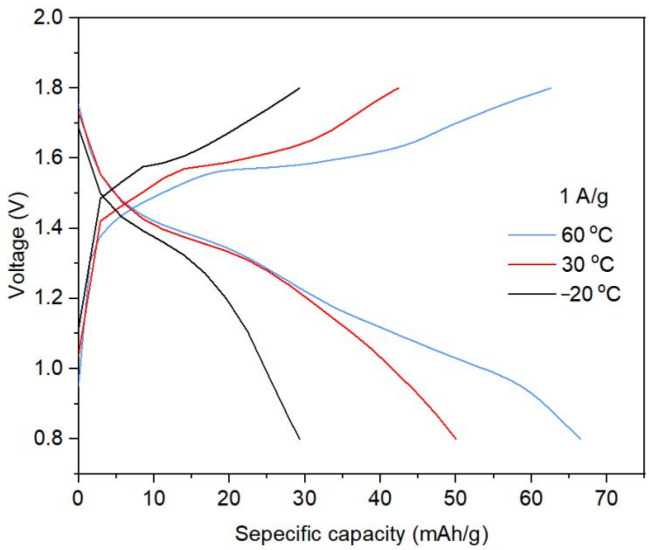


Figure S48. Charge–discharge curves of the Zn|**1**$^{\mathbf{Zn}}$|MnO_2 battery at 1 A/g at different temperature.

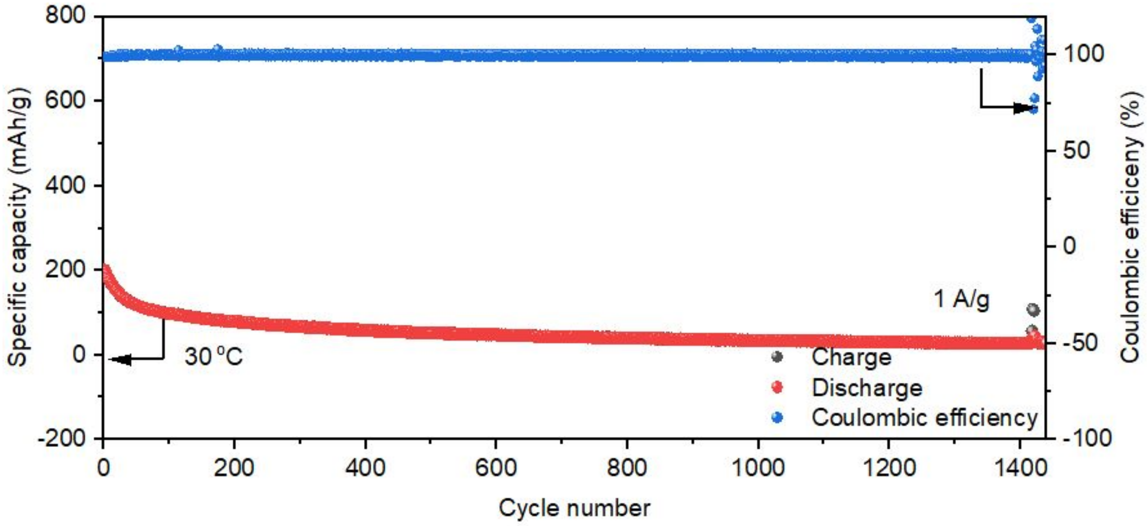


Figure S49. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$| PTCDA battery at 1 A/g at room temperature.

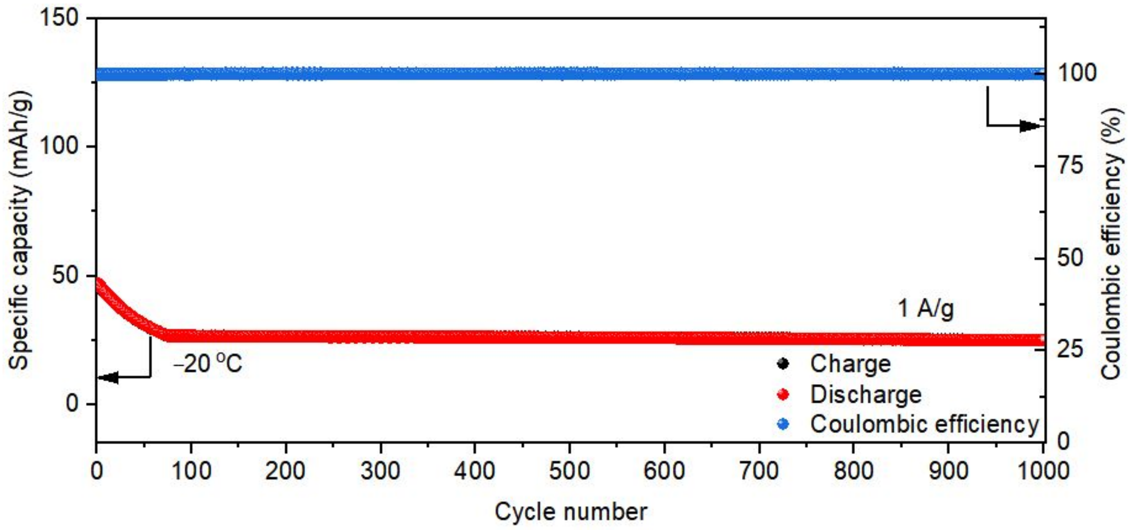


Figure S50. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$| PTCDA battery at 1 A/g at -20 ^{o}C.

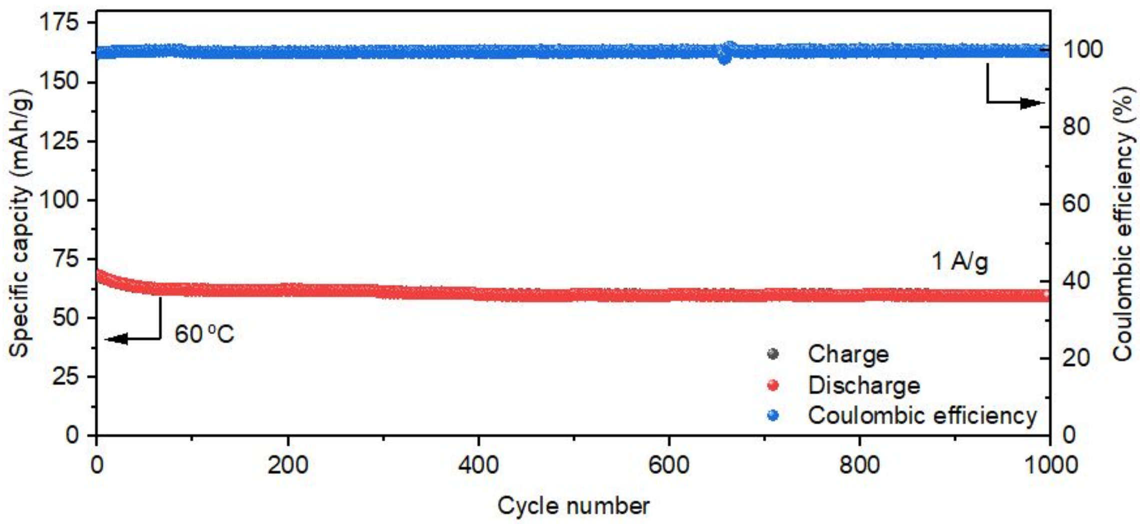


Figure S51. Long-term GCD performance of the Zn|**1**$^{\mathbf{Zn}}$| PTCDA battery at 1 A/g at 60 ^{o}C.

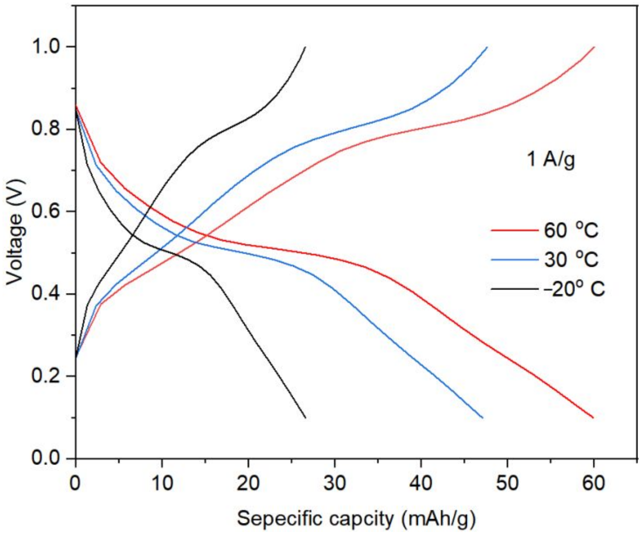


Figure S52. Charge–discharge curves of the Zn|**1**$^{\mathbf{Zn}}$**|**PTCDA battery at 1 A/g at different temperature.

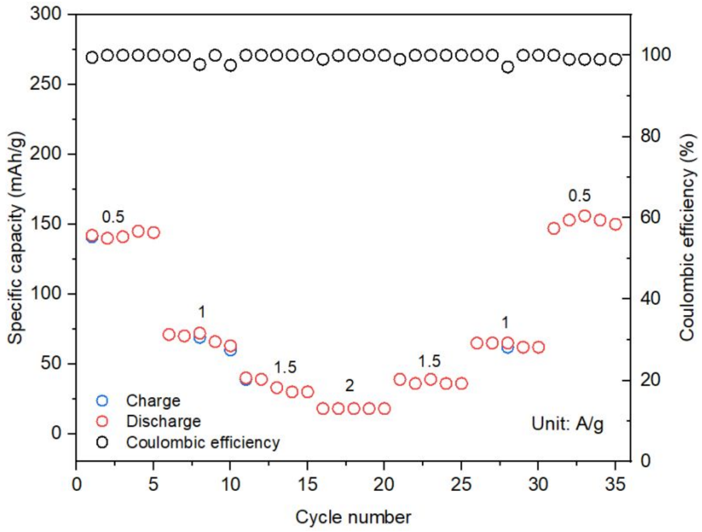


Figure S53. Galvanostatic charge-discharge curves of the Zn|**1**$^{\mathbf{Zn}}$| PTCDA battery at different rates

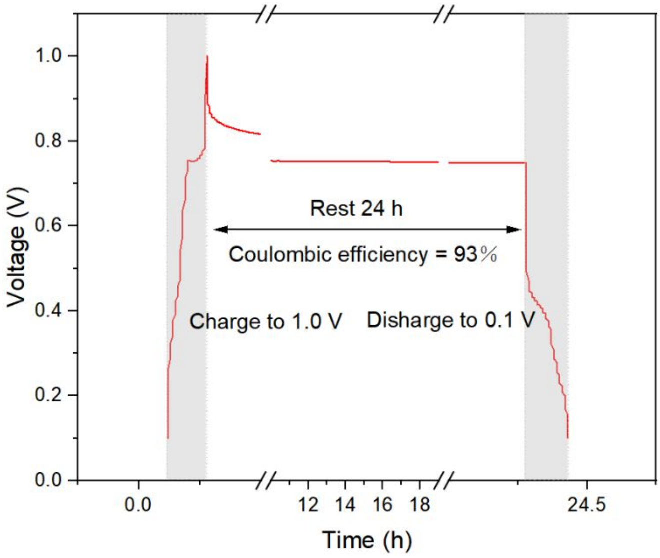


Figure S54. Self-discharge tests of the Zn|**1**$^{\mathbf{Zn}}$| PTCDA battery

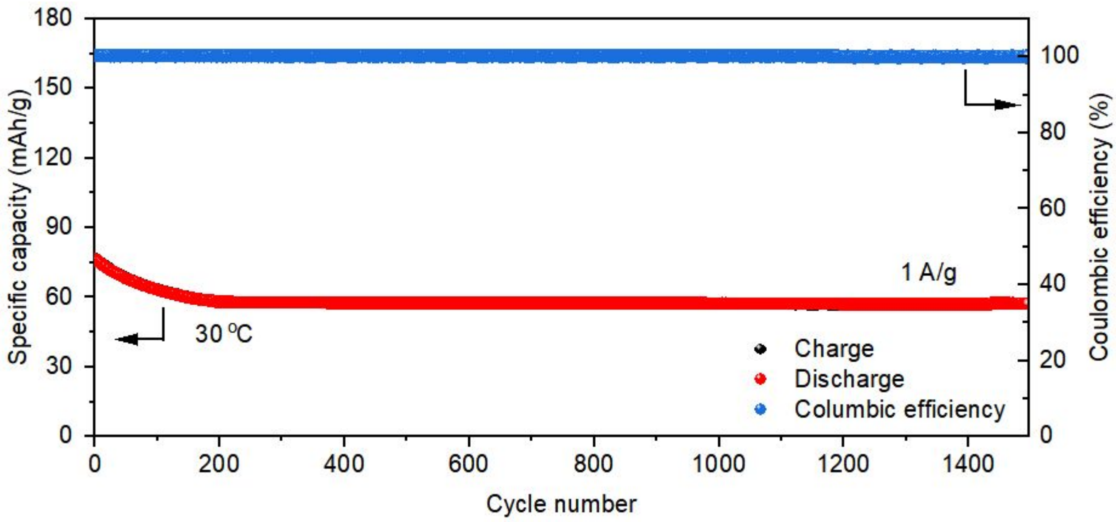


Figure S55. Long-term GCD performance of the Zn|**1**^{Zn}| $Cu_3(HHTP)_2$ battery at 1 A/g at room temperature

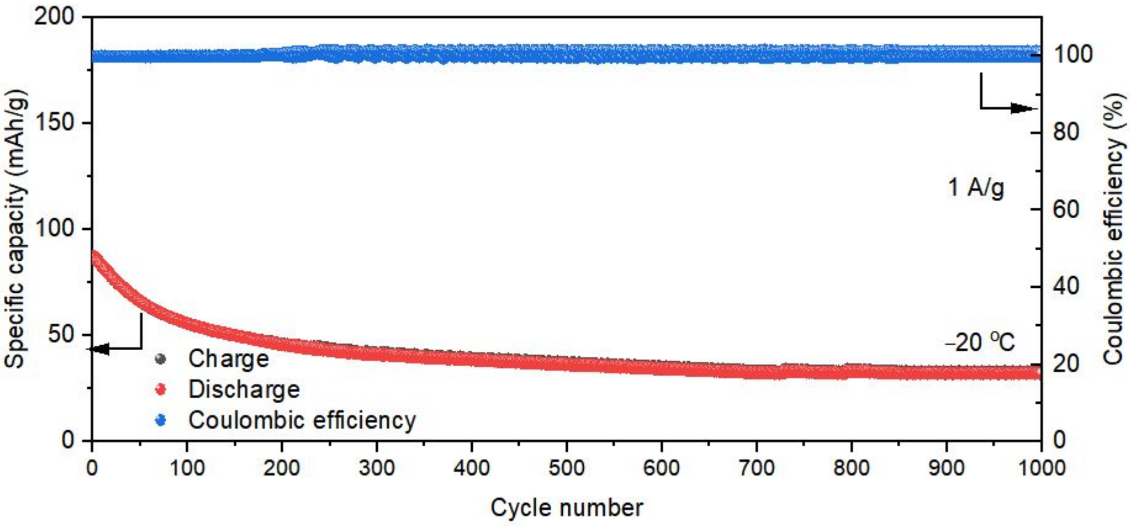


Figure S56. Long-term GCD performance of the Zn|**1**^{Zn}| $Cu_3(HHTP)_2$ battery at 1 A/g at -20 ^{o}C

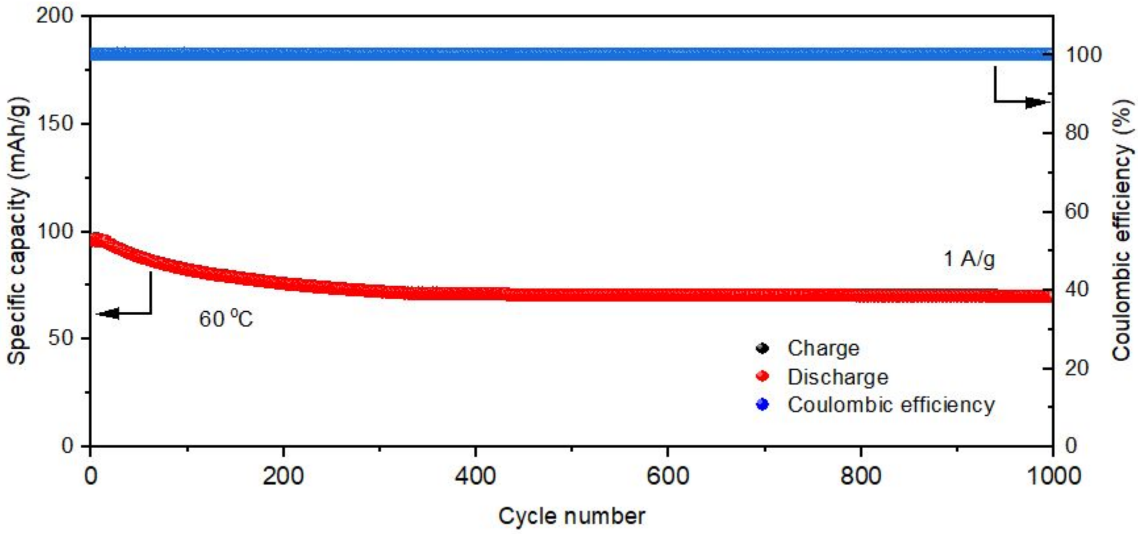


Figure S57. Long-term GCD performance of the Zn|**1**^{Zn}|$Cu_3(HHTP)_2$ battery at 1 A/g at 60 ^{o}C

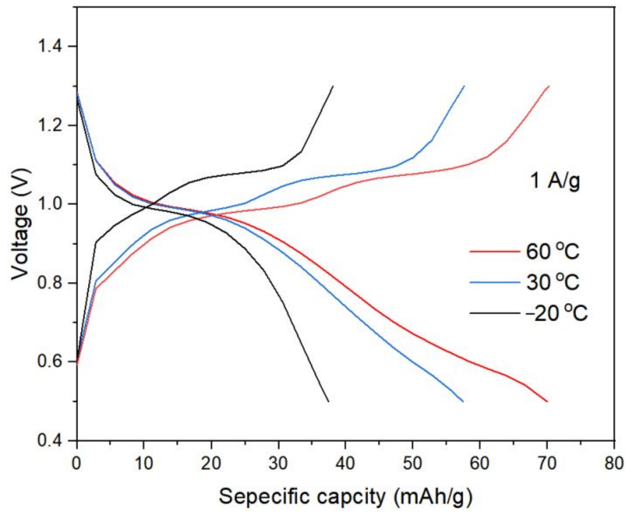


Figure S58. Charge–discharge curves of the Zn|**1**$^{\mathbf{Zn}}$| $Cu_3(HHTP)_2$ battery at 1 A/g at different temperature.

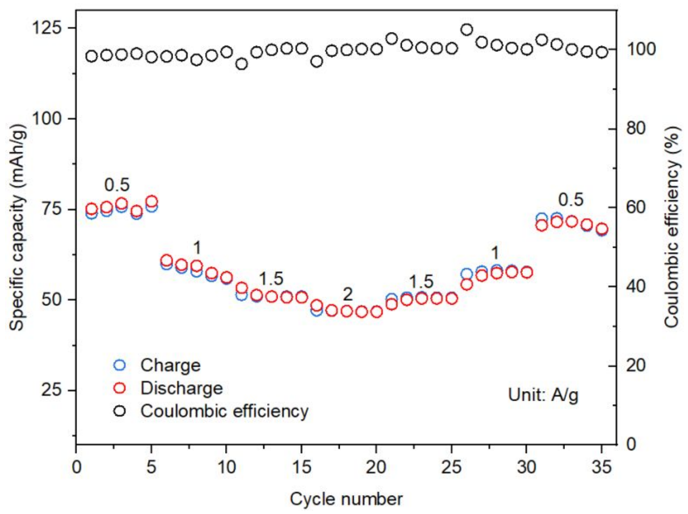


Figure S59. Galvanostatic charge-discharge curves of the Zn|**1**$^{\mathbf{Zn}}$ |$Cu_3(HHTP)_2$ battery at different rates.

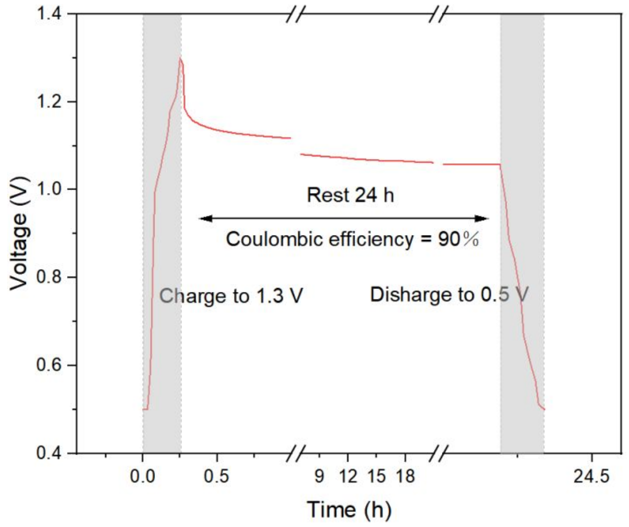


Figure S60. Self-discharge tests of the Zn|$\mathbf{1}^{\mathbf{Zn}}$| $Cu_3(HHTP)_2$ battery.

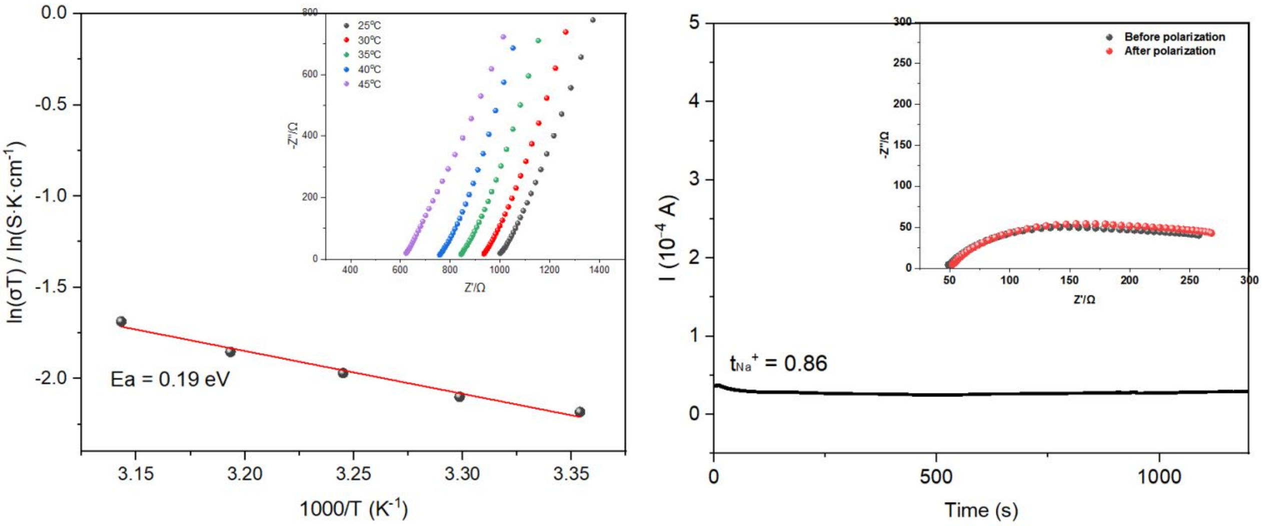


Figure S61. Ionic conductivity and transference number of $\mathbf{1^{Na}}$

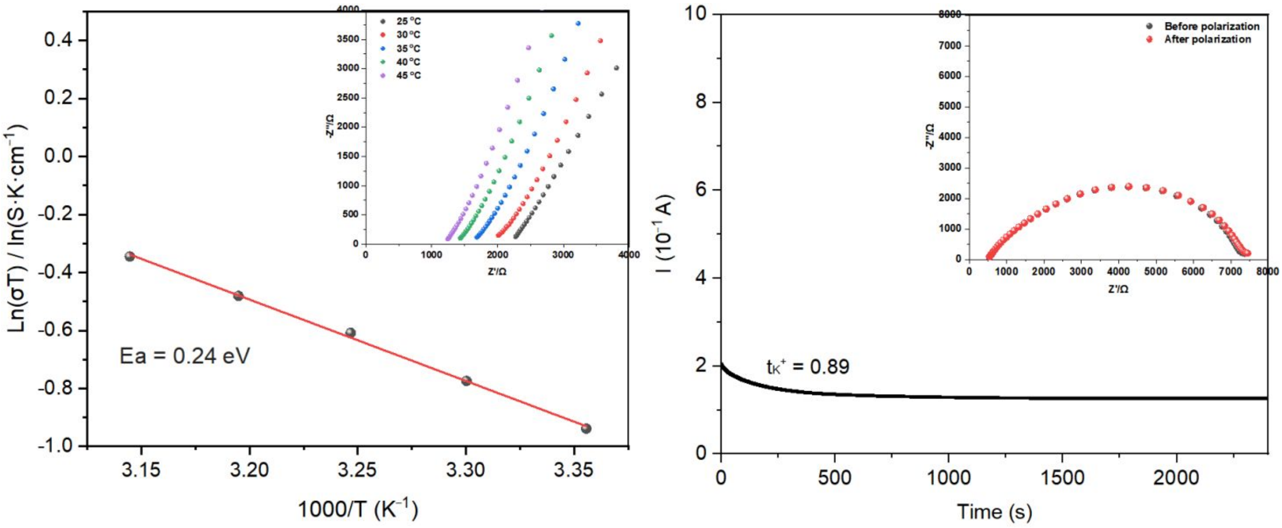


Figure S62. Ionic conductivity and transference number of $\mathbf{1^{K}}$

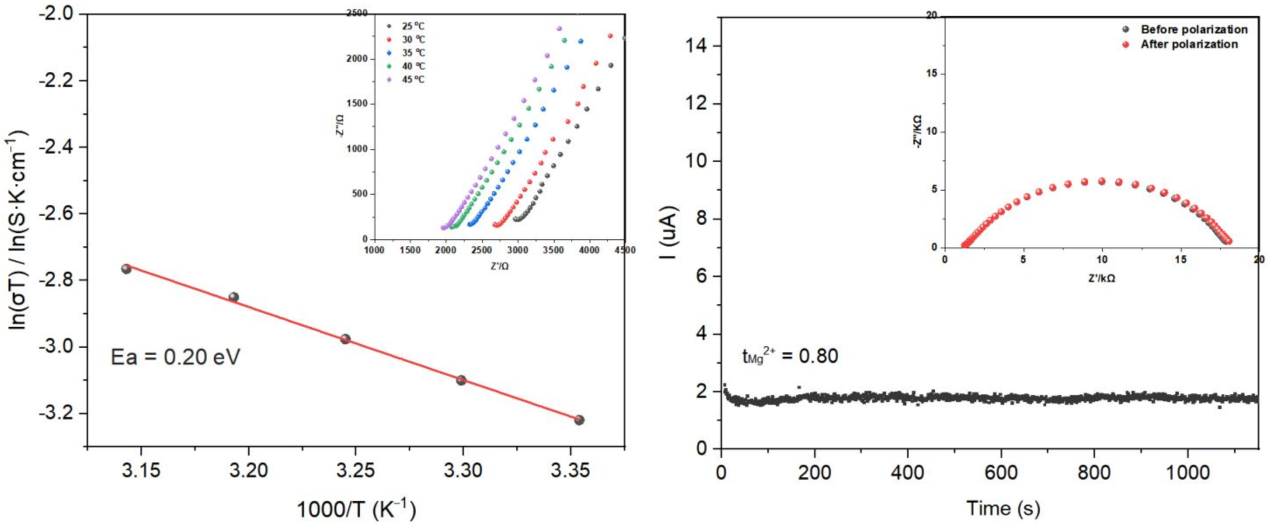


Figure S63. Ionic conductivity and transference number of $\mathbf{1^{Mg}}$

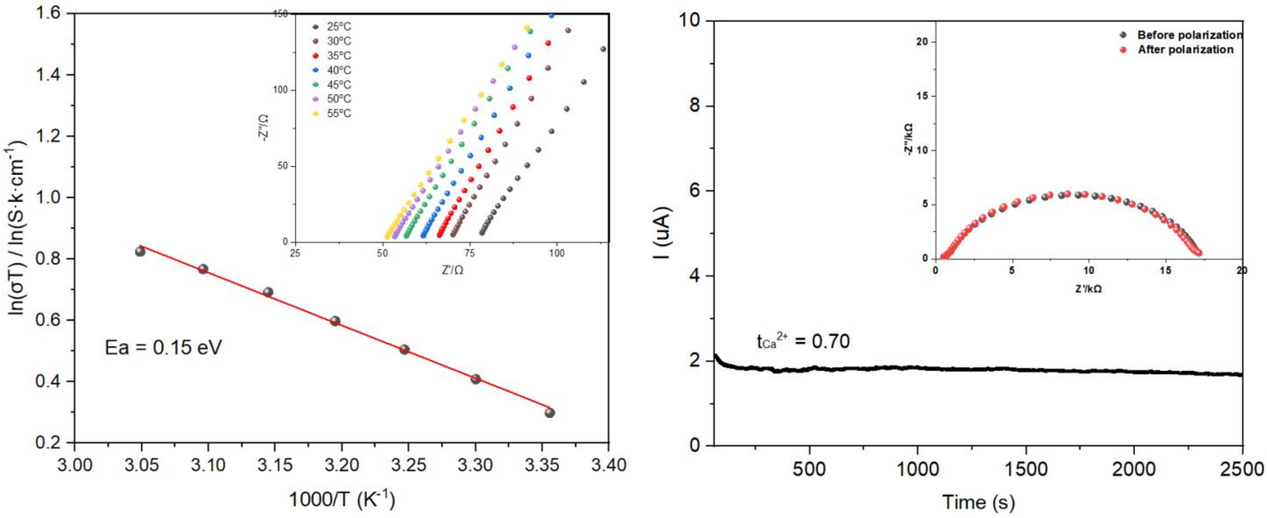


Figure S64. Ionic conductivity and transference number of $\mathbf{1^{Ca}}$

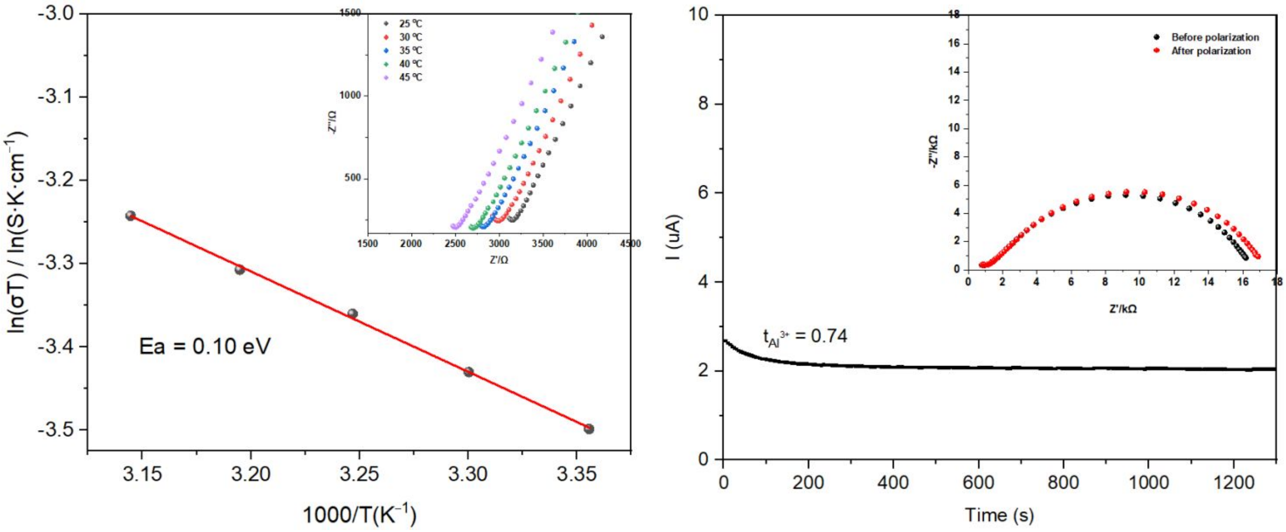


Figure S65. Ionic conductivity and transference number of $\mathbf{1^{Al}}$

Table S2. Ionic Conductivity, activation energy, and transference number of $\mathbf{1^{M}}$ at room temperature.

Electrolyte	σ (S/cm)	E_a (eV)	t
$\mathbf{1^{K}}$	1.71×10^{-4}	0.24	0.89
$\mathbf{1^{Ca}}$	4.46×10^{-3}	0.15	0.70
$\mathbf{1^{Na}}$	3.88×10^{-4}	0.19	0.86
$\mathbf{1^{Mg}}$	1.30×10^{-4}	0.20	0.80
$\mathbf{1^{Al}}$	1.11×10^{-4}	0.10	0.74

1^{Zn}	2.20×10^{-3}	0.19	0.90

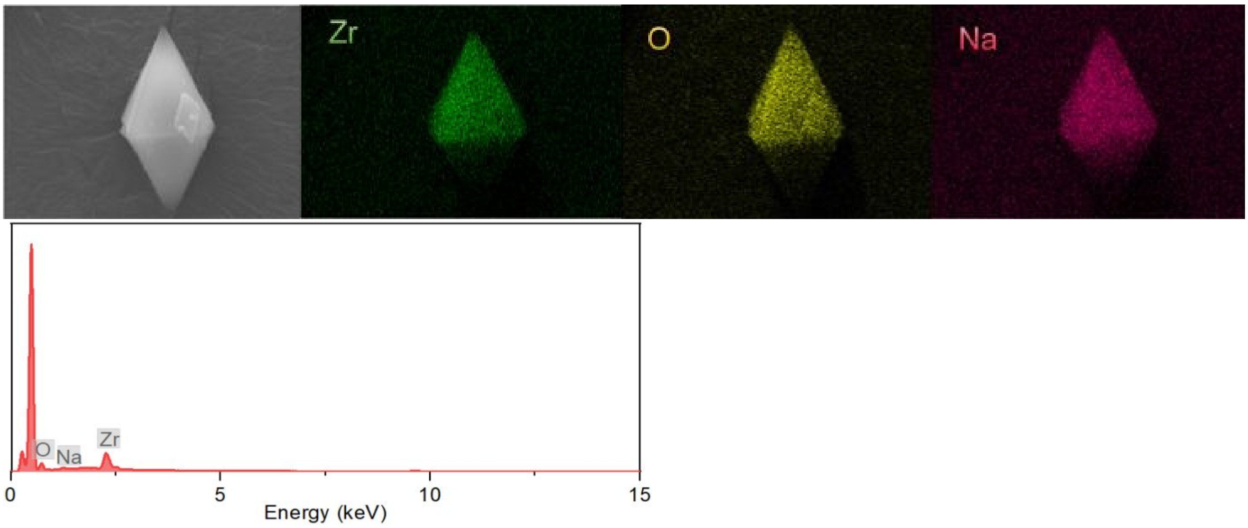


Figure S66. SEM and SEM-EDS mapping of $\mathbf{1}^{Na}$

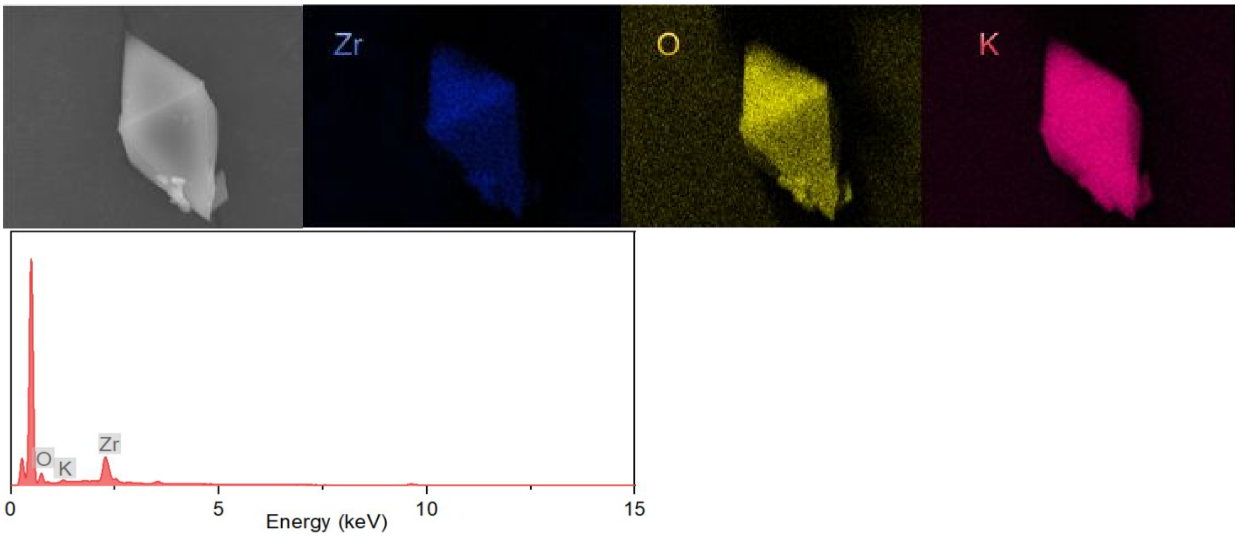


Figure S67. SEM and SEM-EDS mapping of $\mathbf{1}^{K}$

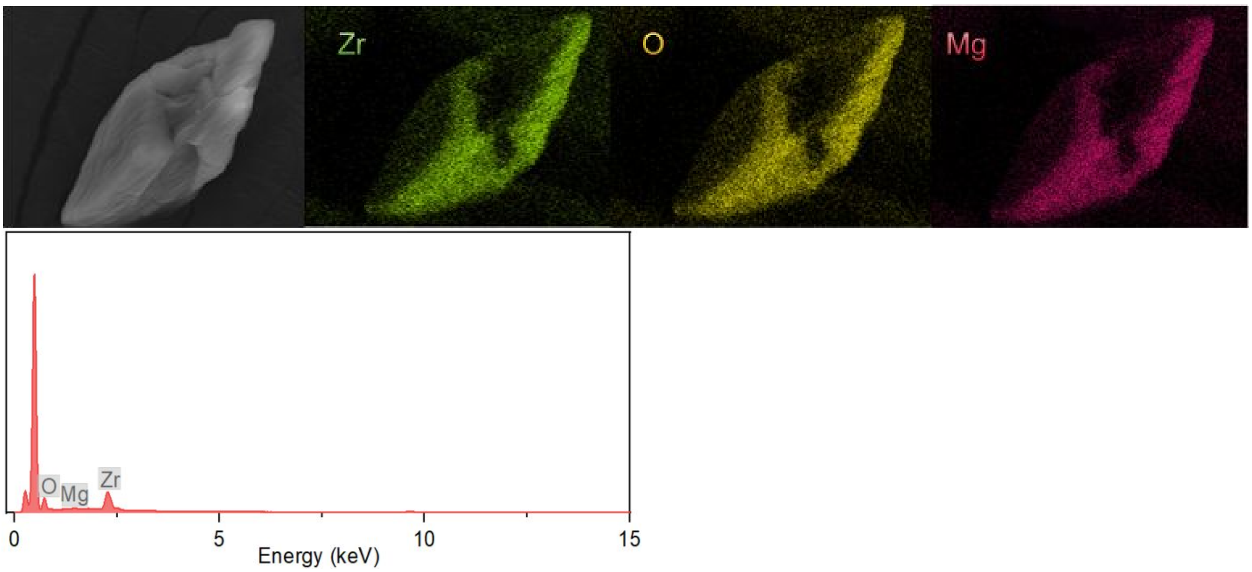


Figure S68. SEM and SEM-EDS mapping of $\mathbf{1}^{Mg}$

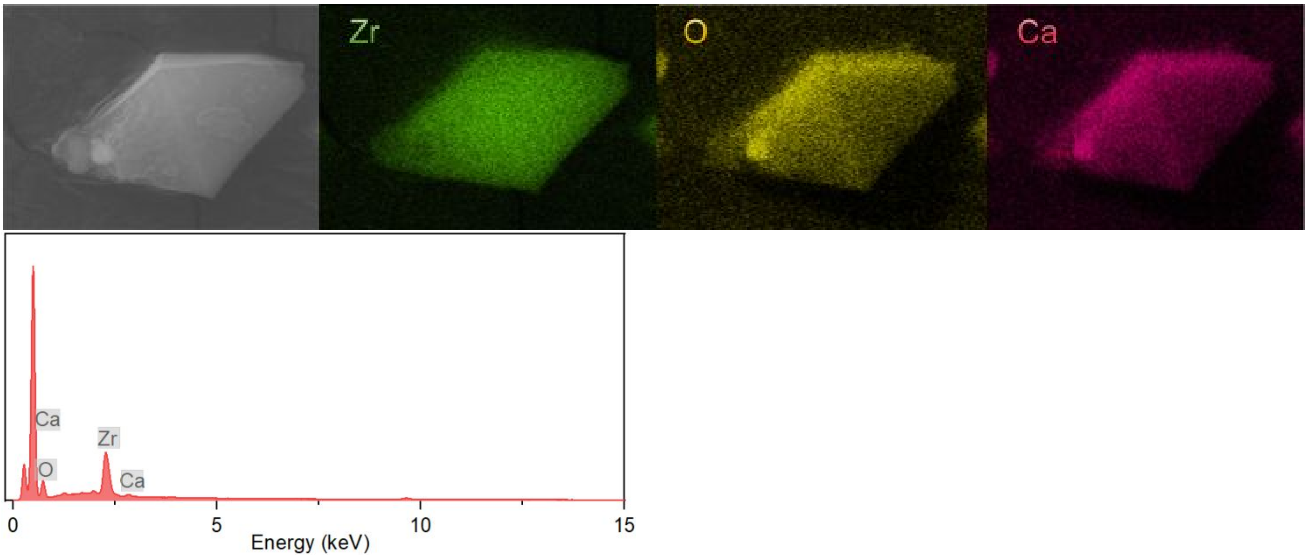


Figure S69. SEM and SEM-EDS mapping of $\mathbf{1}^{\mathbf{Ca}}$

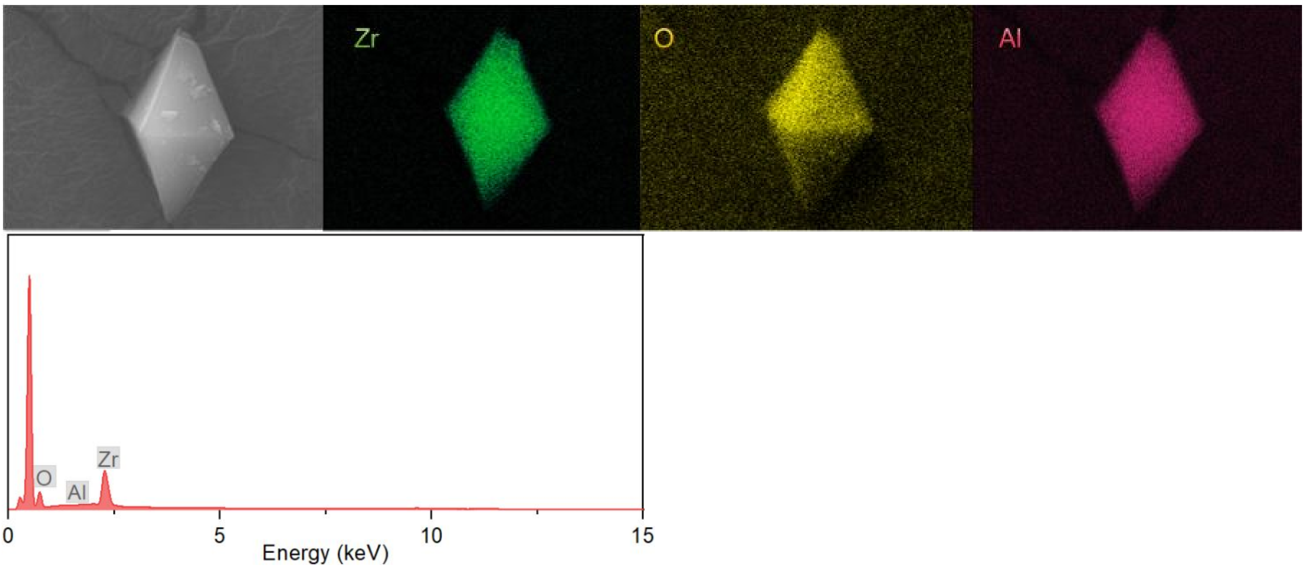


Figure S70. SEM and SEM-EDS mapping of $\mathbf{1}^{\mathbf{Al}}$

5 Summary and Comparison

5.1 Summary of Porous Materials for Solid-State Zn^{2+} Electrolyte

Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^{+}	Ref.
1	TpPa-$SO_3Zn_{0.5}$	2.2×10^{-4}	r.t.	0.19	0.91	*Chem. Sci.*, **2020**, *11*, 11692–11698
2	InOF-Zn	1.22×10^{-3}	r.t.	——	——	*Inorg. Chem.* **2021**, *60*, 11032–11037
3	Zn@MOF-808	2.1×10^{-4}	30	0.12	0.93	*Nano Energy.* **2019**, *56*, 92-99
4	ZnZnBTT	1.15×10^{-4}	r.t.	0.317	——	*J. Am. Chem. Soc.* **2023**, *145*, 25962−25965
5	ZnI_2@TB-MOF	6.10×10^{-4}	r.t.	0.20	0.84	*Chem. Sci.*, **2024**, *15*, 17579-17589
6	**1**^{Zn}	2.20×10^{-3}	r.t.	0.19	0.90	This work

5.2 Summary of Porous Materials for Solid-State Mg^{2+} Electrolyte

Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^{+}	Ref.
1	Mg-MOF-74 ⊃ {Mg(TFSI)2}0.15	2.6×10^{-4}	25	0.26	0.47	*J. Phys. Chem. C* **2021**, *125*, 21124−21130
2	MIL-101⊃{Mg(TFSI)2}1.6	1.9×10^{-3}	25	0.18	0.41	*J. Am. Chem. Soc.* **2022**, *144*, 8669−8675
3	MIT-20-Mg	8.8×10^{-7}	r.t.	0.37	——	*J. Am. Chem. Soc.* **2017**, *139*, 13260−13263
4	MOF-$MgCl_2$	1.2×10^{-5}	25	0.32	——	*J. Am. Chem. Soc.* **2019**, *141*, 4422−4427
	MOF-$MgBr_2$	1.3×10^{-4}		0.24	——	
5	$Mg(OPhCF_3)_2 \cdot Mg(TFSI)_2 \subset Mg_2$(dobpdc)	2.5×10^{-4}	r.t.	0.11	——	*Energy Environ. Sci.*, **2014**, *7*, 667–671
6	MgI_2@TB-MOF	3.28×10^{-4}	r.t.	0.21	0.76	*Chem. Sci.*, **2024**, *15*, 17579-17589
7	**1**^{Mg}	1.30×10^{-4}	r.t.	0.20	0.80	This work

5.3 Summary of Porous Materials for Solid-State K^{+} Electrolyte

Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^{+}	Ref.
1	InOF-K	7.69×10^{-4}	25	——	——	*Inorg. Chem.* **2021**, *60*, 11032–11037
2	MOF-808-SO_3K	3.1×10^{-5}	30	0.32	0.76	*ACS Appl. Energy Mater.* **2022**, *5*, 8573–8580
3	KI@TB-MOF	1.64×10^{-4}	r.t.	0.25	0.78	*Chem. Sci.*, **2024**, *15*, 17579-17589
4	**1**^{K}	1.71×10^{-4}	r.t.	0.24	0.89	This work

5.4 Summary of Porous Materials for Solid-State Na^+ Electrolyte

Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^+	Ref.
1	MIL-121/Na+SE	1.2×10^{-4}	30	0.36	——	*Adv. Energy Mater.* **2021**, *11*, 2003542
2	InOF-Na	7.97×10^{-4}	25	——	——	*Inorg. Chem.* **2021**, *60*, 11032–11037
3	UIOSNa	3.6×10^{-4}	r.t.	——	0.34	*ACS Appl. Mater. Interfaces* **2021**, *13*, 24662−24669
4	NaOOC-COF	2.68×10^{-4}	r.t.	0.24	0.9	*Nano Energy.*, **2022**, *92*, 106756
5	EHU1(Sc,Na)	9.2×10^{-5}	r.t.	0.64	——	*Chem. Mater.* **2016**, *28*, 2519−2528
6	MIL-121/Na	1.2×10^{-4}	30	0.36	——	*Adv. Energy Mater.* **2021**, 2003542
7	MIT-20-Na	1.8×10^{-5}	r.t.	0.39	——	*J. Am. Chem. Soc.* **2017**, *139*, 13260−13263
8	NaI@TB-MOF	3.09×10^{-4}	r.t.	0.19	0.69	*Chem. Sci.*, **2024**, *15*, 17579-17589
9	**1^{Na}**	3.88×10^{-4}		0.19	0.86	This work

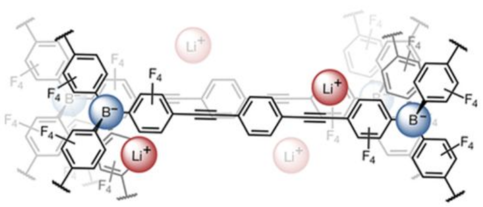
5.5 Summary of Porous Materials for Solid-State Li^+ Electrolyte

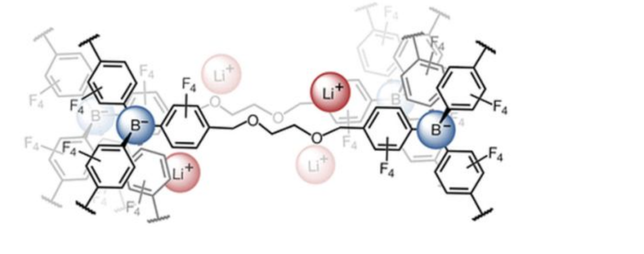
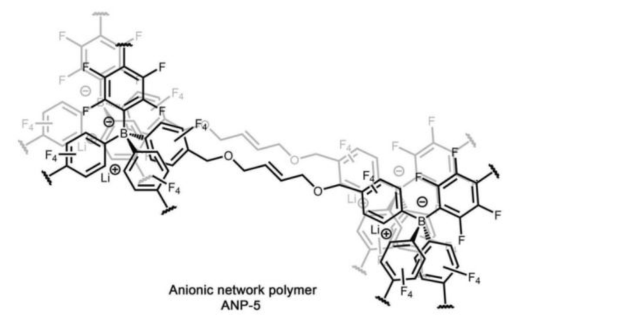
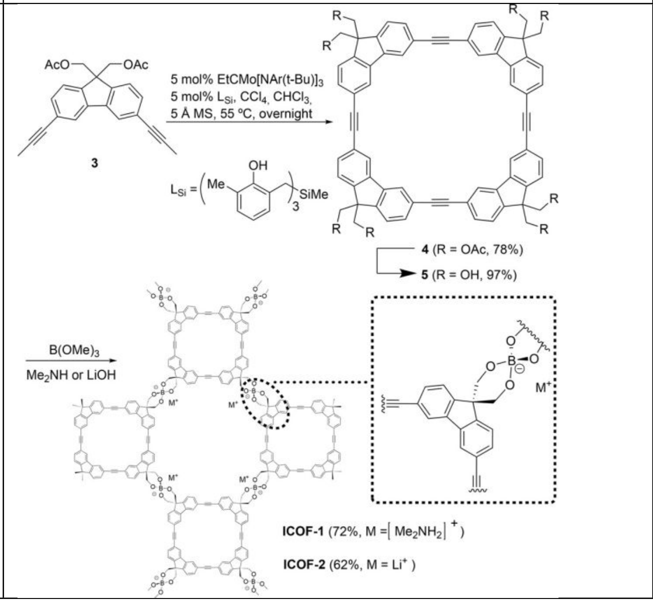
Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^+	Ref.
1	IL/LiTFSI@NUST-9	2.60×10^{-3}	120	0.317	——	*Chem. Mater.* **2021**, *33*, 5058−5066
2	Li^+-PEG@NUST-23	1.36×10^{-3}	80	——	——	*Chem. Mater.* **2022**, *34*, 9104−9110
3	COF-PEGB6-Li	1.5×10^{-3}	200	0.60	0.3	*ACS Appl. Energy Mater.* **2021**, *4*, 11720−11725
4	UiO-66-S-Li@OPBI	1.46×10^{-3}	r.t	——	0.73	*ACS Appl. Energy Mater.* **2022**, *5*, 9131−9140
5	Helical Covalent Polymer	1.2×10^{-3}	r.t.	0.14	0.84	*CCS Chem.* **2021**, *3*, 2762–2770
6	NH_2-MIL-125-Li	1.52×10^{-4}	r.t.	0.15	0.72	*J. Am. Chem. Soc.* **2023**, *145*, 5718−5729
	MIL-125-Li	4.42×10^{-5}		0.24	0.43	
7	COF-MCMCs	4.9×10^{-4}	30	0.31	0.71	*Angew. Chem. Int. Ed.* **2023**, e202302505
8	Dq-COF-DEME	1.07×10^{-3}	r.t.	——	0.382	*Electrochim. Acta.* **2021**, *389*,138585
	EB-COF-TFSI-DEME	1.23×10^{-3}		——	0.396	
9	EHU1(Sc,Li)	4.2×10^{-4}	r.t	0.25	**---**	*Chem. Mater.* **2016**, *28*, 2519−2528
10	UiO-66-LiSS	7.8×10^{-4}	r.t.	0.21	0.90	*ACS Appl. Energy Mater.* **2020**, *3*, 4007−4013
11	C−CSE	4.26×10^{-4}	30	——	0.67	*ACS Appl. Mater. Interfaces* **2020**, *12*, 22930−22938
12	Li/UiOLiTFSI	2.07×10^{-4}	25	0.31	0.84	*ACS Appl. Mater. Interfaces* **2019**, *11*, 43206–43213
13	Li-IL@MOF	3.0×10^{-4}	r.t.	——	0.36	*Adv. Mater.* **2018**, *30*, 1704436
14	UiO-67-Li	6.4×10^{-4}	r.t.	0.14	0.65	*Adv. Funct. Mater.* **2022**, 2113235
15	DMA@LiTFSI-mediated COF	1.7×10^{-4}	r.t.	——	0.85	*Adv. Mater.* **2022**, *34*, 2201410

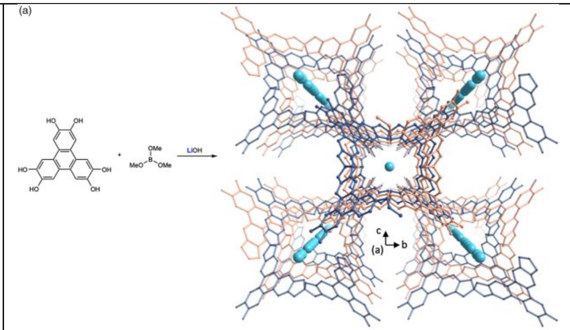
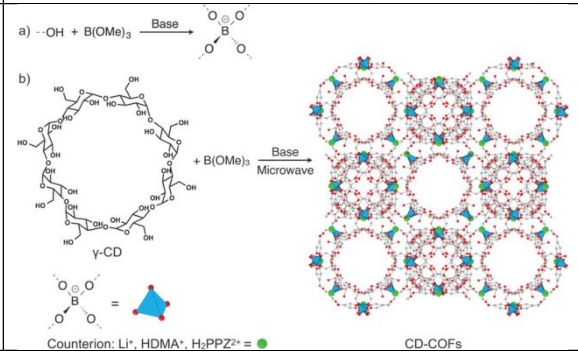
16	PSS@HKUST-1	5.53×10^{-4}	25	0.21	—	*Angew. Chem. Int. Ed.* **2016**, *55*, 15120–15124
17	CD-COFs	2.7×10^{-3}	30	0.26	—	*Angew. Chem. Int. Ed.* **2017**, *56*, 16313–16317
18	PEG-$LiBF_4$ @DBC-2P (COF)	2.31×10^{-3}	70	0.09	—	*Angew. Chem. Int. Ed.* **2019**, *58*, 15742–15746
19	$LiPF_6$@PAF-1	4.0×10^{-4}	r.t.	0.15	0.859	*Angew. Chem. Int. Ed.* **2020**, *59*, 769–774
20	Q-COF SSE	1.5×10^{-4}	60	0.15	0.72	*Angew. Chem. Int. Ed.* **2021**, *60,* 24915–24923
21	ICOF-2	3.05×10^{-4}	r.t.	0.24	0.80	*Angew. Chem. Int. Ed.* **2016**, *55,* 1737–1741
22	LiOtBu-grafted UiO-66	1.8×10^{-5}	r.t.	0.18	—	*Chem. Eur. J.* **2013**, *19,* 5533–5536
23	$LiPF_6$/Ge-COF-1	2.5×10^{-4}	30	0.29	0.67	*Chem. Eur. J.* **2019**, *25*, 13479–13483
24	DMPI @ Tp-DB-COF	4.36×10^{-4}	60	0.21	0.58	*Chem. Eur. J.* **2021**, *27*, 4583–4587
25	Tp-PaSO_3Li-COF	1.6×10^{-3}	20	0.13	0.94	*Chem. Commun.*, **2020**, *56*, 2747–2750
26	0.2MOF-LPC	1.3×10^{-3}	r.t	0.15	0.58	*Chem. Commun.*, **2020,** *56*, 13603–13606
	0.2MOF-LPF	2.4×10^{-3}		0.11	0.74	
	0.2MOF-LFS	1.0×10^{-3}		0.08	0.55	
27	$LiPF_6$-LE@COF	4.63×10^{-3}	20	5.2 kJ/mol	0.79	*Chem. Commun.*, **2020**, *56*, 10465–10468
28	Li-AOIA@BF_4 (MOF)	1.09×10^{-5}	r.t.	0.18	—	*Chem. Commun.*, **2020**, *56*, 14873–14876
29	ZIF-67@ZIF-8	1.35×10^{-3}	r.t.	0.18	0.82	*Chem. Commun.*, **2020**, *56*, 14629–14632
30	VDF-HFP/MOF-5-II	1.2×10^{-3}	25	—	0.9	*Chem. Commun.*, **2022**, *58*, 6717–6720
31	$LiNO_3$@HM-MIL-101	1.24×10^{-3}	30	0.11	0.86	*J. Mater. Chem. A*. **2022**, *10*, 14020–14027
32	Mg_2(dobdc)·0.35LiO^{i}Pr·0.25LiBF_4 ·EC·DEC	3.1×10^{-4}	27	0.15	—	*J. Am. Chem. Soc.* **2011**, *133*, 14522–14525
33	$LiClO_4$@COF-5	2.6×10^{-4}	r.t.	37±4 meV	—	*J. Am. Chem. Soc.* **2016**, *138*, 9767–9770
	$LiClO_4$@TpPa-1	1.5×10^{-4}		—	—	
34	MIT-20-LiBF_4	4.8×10^{-4}	r.t.	0.16	0.66	*J. Am. Chem. Soc.* **2017**, *139*, 13260−13263
35	Li-CON-TFSI	2.09×10^{-4}	70	0.34	0.61	*J. Am. Chem. Soc.* **2018**, *140*, 896−899
36	SLEN material	1.0×10^{-3}	r.t.	0.16	0.56	*J. Am. Chem. Soc.* **2018**, *140*, 7504−7509
37	Li^+@TPB-DMTP-COF	1.36×10^{-7}	40	0.96	—	*J. Am. Chem. Soc.* **2018**, *140*, 7429−7432
	Li^+@TPB-BMTP-COF	5.49×10^{-4}	90	0.87	—	
38	COF-PEO-3-Li	9.72×10^{-5}	200	—	—	*J. Am. Chem. Soc.* **2019**, *141*, 1227−1234
	COF-PEO-6-Li	3.71×10^{-4}		—	—	
	COF-PEO-9-Li	1.33×10^{-4}		—	—	
39	MOF-LiCl	2.4×10^{-5}	25	0.34	0.69	*J. Am. Chem. Soc.* **2019,** *141*, 4422−4427
	MOF-LiBr	3.2×10^{-5}		0.30	0.42	
	MOF-LiI	1.1×10^{-4}		0.24	0.34	
40	PEG-Li^+@CD-COF-Li	1.30×10^{-4}	120	0.17	—	*J. Am. Chem. Soc.* **2019**, *141*, 1923−1927
	PEG-Li^+@COF-300	9.11×10^{-5}		0.20	—	
	PEG-Li^+@COF-5	3.49×10^{-5}		0.35	—	
	PEG-Li^+@EB-COF-ClO_4	1.78×10^{-3}		0.21	—	

41	COF-TpPa-SO_3Li	2.7×10^{-5}	r.t.	0.18	0.90	*J. Am. Chem. Soc.* **2019**, *141*, 5880−5885
42	CH_3-ImCOF	8.0×10^{-5}	r.t.	0.27	0.93	*J. Am. Chem. Soc.* **2019**, *141*, 7518−7525
	H-ImCOF	5.3×10^{-3}		0.12	0.88	
	CF_3-ImCOF	7.2×10^{-3}		0.10	0.81	
43	MOF-688	3.4×10^{-4}	20	——	0.87	*J. Am. Chem. Soc.* **2019**, *141*, 17522−17526
44	UIO/Li-IL SEs	3.2×10^{-4}	25	0.4	0.33	*Small*. **2019**, *15*, 1804413
45	InOF-Li	1.49×10^{-3}	25	0.19	0.78	*Inorg. Chem.* **2021**, *60*, 11032–11037
46	MIL-121/Li+SE	9.1×10^{-3}	50	0.28	——	*Adv. Energy Mater.* **2021**, *11*, 2003542
47	LPC@HKUST-1	3.8×10^{-4}	r.t.	0.18	——	*Adv. Mater.* **2018**, 1707476
	LPC@MIL-100-Al	1.22×10^{-3}		0.21	——	
	LPC@MIL-100-Fe	9×10^{-4}		0.18	——	
	LPC@UiO-67	6.5×10^{-4}		0.12	0.65	
	LPC@UiO-66	1.8×10^{-4}		0.21	——	
	LPC@MIL-100-Cr	2.3×10^{-4}		0.18	——	
49	Al-Td-MOF-1	$5.7\pm1.9 \times 10^{-4}$	r.t.	0.11	——	*Angew. Chem. Int. Ed.* **2018**, *57*, 16683 –16687
50	LiI@TB-MOF	2.75×10^{-3}	r.t.	0.15	0.89	*Chem. Sci.*, **2024**, *15*, 17579-17589
	LiBr@TB-MOF	1.33×10^{-3}		0.19	0.91	
	LiCl@TB-MOF	1.34×10^{-4}		0.21	0.89	
	$LiClO_4$@TB-MOF	1.51×10^{-4}		0.22	0.80	
	LiOTf@TB-MOF	2.66×10^{-5}		0.22	0.74	

5.6 Summary of Anionic Boron-Based Porous Materials for Solid-State Electrolyte

Entry	Compound	σ (S/cm)	T (°C)	Ea (eV)	t^+	Ref.
1		2.1×10^{-4}	28	0.25	0.93	*Chem. Sci.*, **2015**, *6*, 5499–5505

2		3.5×10^{-4}	25	0.22	1.01	*Macromolecules* **2021**, *54*, 7582−7589
3		1.5×10^{-4}	26	0.24	0.95	*Adv. Mater.* **2020**, *32*, 1905771
4		3.05×10^{-4} (ICOF-2)	r.t.	0.24	0.80	*Angew. Chem. Int. Ed.* **2016**, *55*, 1737–1741

5		1.2×10^{-3}	r.t.	0.14	0.84	*CCS Chem.* **2021**, *3*, 2762–2770
6		2.7×10^{-3}	30	0.26	—	*Angew. Chem. Int. Ed.* **2017**, *56*, 16313–16317

5.7 Summary of Anionic MOFs and COFs as Solid Electrolyte for Solid-State Batteries

Compound	σ (mS/cm)	Ea (eV)	t^+	Full cell					Ref.
				Type	Cycle	Current Density (mA/ cm^2)	Capacity (mAh/g)	Capacity Retention	
COF-SS-Li	0.0963	0.24	0.53	LFP/Li	500	0.1 A/g	120	56.7%	*ACS Appl. Mater. Interfaces* **2023**, *15*, 34704−34710
COF-$TFSI^{-}Li^{+}$	0.0826	0.14	0.91	LFP/Li	200	0.1C	142	94.5%	*ACS Energy Lett.* **2024**, *9*, 5341−5348
Li^{+}[Cu-BTC]$^{-}$	0.42	0.18	0.8	LFP/Li	150	0.1C	152	82%	*Chem. Commun.*, **2024**, *60*, 13416 – 13419
COF-S-S-TFSILi	0.43	0.24	0.90	LFP/Li	100	0.2C	150.3	98.8%	*Polym. Chem.*, **2024**, *15*, 4529–4541
MOF-688	0.34	——	0.87	LFP/Li	200	30 (mA/g)	149	83%	*J. Am. Chem. Soc.* **2019**, *141*, 17522−17526
IL-1.0@NUST-7	0.087	0.32	—	LFP/Li	60	0.05C	140.8	93%	*Chem. Mater.* **2021**, *33*, 5058−5066
LiI@TB-MOF	2.75×10^{-3}	0.15	0.89	LFP/Li	220	0.5	135	95%	*Chem. Sci.*, **2024**, *15*, 17579-17589
CN-iCOF	0.0133	0.2	0.93	LCO/Li	130	0.05C	160	63%	*Angew. Chem. Int. Ed.* **2024**, *63*, e202410392
PEO/5%TpPa-SO_3Li	0.628 (60^{o}C)	——	0.78	S/Li	600 (60 ^{o}C)	0.5C (60 ^{o}C)	506	——	*J. Colloid Interface Sci.***2025**, *678*, 105-113
QSSE-COF-SO_3Na	0.41	0.14	0.89	NVOPF /Na	700	1C	92	82%	*Energy Storage Mater.* **2025**, *77*, 104192
PEGMEM-co-SSS@ZIF-8	0.4 (80 ^{o}C)	——	0.87	NVP/Na	200	0.1C (80 ^{o}C)	92	95%	*Angew. Chem. Int. Ed.* **2024**, *63*, e202318822
Na-MOFs Mg-MOFs Al-MOFs	0.44 1.0 0.13	0.28 0.20 0.37	—	NVP/Na	200	0.5C	~90	>95%	*ACS Appl. Mater. Interfaces* **2020**, *12*, 43824−43832
TpPa-$SO_3Zn_{0.5}$	0.22	0.19	0.91	α-MnO_2/Zn	800	0.6 A/g	196	73%	*Chem. Sci.*, **2020**, *11*, 11692–11698
TCOF-S-Gel	14.1	——	0.89	α-MnO_2/Zn	1400	2C	~200	——	*Angew. Chem. Int. Ed.* **2023**, *62*, e202312020
1^{Zn}	2.2	0.19	0.90	NVO/Zn	2500	5A/g	351	99.7%	Our work

5.8 Summary of other Anionic Solid Electrolytes for Solid-State Batteries

Compound	σ (mS/cm)	Ea (eV)	t^+	Full cell					
				Type	Cycle	Current Density (mA/ cm^2)	Capacity (mAh/g)	Capacity Retention	
PAEK-SIGPE	0.078 (30 °C)	0.12	0.95	LFP/Li	300	0.2C	117	88%	*Appl. Surf. Sci.* **2023**, *611*, 155363
VC70	0.16	——	0.92	LFP/Li	300	0.1C	150	60%	*J. Power Sources.* **2023**, *574*, 233145
M-PBI/PEO-0.4	0.099	——	0.89	LFP/Li	200	0.2C	152.8	74%	*Chem. Eng. J.* **2025**, *503*, 158268
6FPSF-Li	0.246 (40 °C)	——	0.93	LFP/Li	250	0.1C	157	83%	*J. Energy Chem.* **2023**, *80*, 174-181
PLSSCQD	0.202	——	0.94	LFP/Li	100	0.2 C (60 °C)	155.9	94.3%	*Nano Energy,* **2021**, *82*, 105698
ANP-5	0.48 (40 °C)	0.16	0.95	LFP/Li	50	0.5C	122	——	*Adv. Mater.* **2020**, *32*, 1905771
C-ASPE-5/8	0.0196	——	0.91	LFP/Li	100	0.1C	137.7	89%	*Macromolecules* **2021**, *54*, 9135−9144
MCM41-NLiTf	0.024	——	0.82	LFP/Li	100	0.1C	~140	96.9%	*Chem. Commun.*, **2022**, *58*, 13656 - 13659
PI-SIGPE	0.14	0.10	0.97	LFP/Li	330	0.2C	145	90.1%	*J. Mater. Chem. A,* **2023**, *11*, 1766–1773
IN-SCPE5	0.19	——	0.9	LFP/Li	100	0.2C	150.2	92.5%	*ACS Appl. Energy Mater.* **2020**, *3*, 12532−12539
SLIC-PEM5	0.18	——	0.91	LFP/Li	100	0.2C	153	92%	*ACS Appl. Energy Mater.* **2019**, *2*, 3028−3034
PAF-220-QSPE	0.206	0.14	0.76	LFP/Li	180	0.2C	170	90%	*Small.* **2023**, *19*, 2302818
c-SIPM60	0.12	—	0.87	LFP/Li	700	1C	134.9	75.6%	*ACS Appl. Mater. Interfaces* **2025**, *17*, 5360−5369
PEO_8−LiPCSI	0.073(60 °C)	——	0.84	LFP/Li	80	0.1C	141	85%	*ACS Appl. Mater. Interfaces* **2020**, *12*, 7249−7256
Li^+@iPOPs	4.38	0.63	0.95	LFP/Li	200	0.5C	114	86.7	*ACS Appl. Mater. Interfaces* **2024**, *16*, 44957−44966
SI10-05-70%PC	0.6	——	——	NCM811/Li	180	1C	165	97.2%	*Nano Energy,* **2020**, *77*, 105129
AMPE	0.38	0.24	——	LCO/Li	700	0.2C	152.08	——	*Angew. Chem. Int. Ed.* **2024**, *63*, e202412280
NaSTFSI-SIPEs	0.042	0.12	—	PW/Na	200	0.4C (40 °C)	102	69%	*ACS Appl. Polym. Mater.* **2025**, *7*, 4895−4907
SIHE	2.15	——	0.93	V_2O_5/Zn	500	5C	127.5	81.9%	*ACS Appl. Mater. Interfaces* **2021**, *13*, 30594−30602
1^{Zn}	2.2	0.19	0.90	NVO/Zn	2500	5A/g	351	99.7%	Our work

5.9 Summary of Cathodes for Zinc Batteries

Cathode	Working capacity (mA h/g)	Working current density (A/g)	Ref.
VO_2-D	420.8	0.1	*Adv. Mater.* **2024**, *36*, 2400888
V_2O_3	625	0.1	*ACS Nano* **2020**, *14*, 7328–7337
V_2O_5	358.7	0.2	*Chem. Mater.* **2020**, *32*, 4054–4064
	326	0.02	*ACS Appl. Mater. Interfaces* **2020**, *12*, 15305–15312
	357	2	*ACS Appl. Energy Mater.* **2020**, *3*, 11183–11192
	350	0.1	*ACS Appl. Mater. Interfaces* **2023**, *15,* 20089–20099
	473	0.1	*Adv. Mater.* **2023**, *13*, 2301480
La-V_2O_5	439	0.1	*Adv. Mater.* **2023**, *35*, 2301996
$(NH_4)_{0.5}V_2O_5\cdot 0.5H_2O$	490	0.3	*Chem. Sci.*, **2024**,*15*, 2601-2611
$Zn_{0.52}V_2O_{5-a}\cdot 1.8\ H_2O$	286.2	0.2	*Angew. Chem. Int. Ed.* **2022**, *61*, e202207779
$Cu_{0.3}V_2O_5$	315	0.02	*ACS Appl. Energy Mater.* **2021**, *4*, 10197–10202
$Ca_{0.24}V_2O_5\cdot 0.83\ H_2O$	340	0.2C	*Angew. Chem. Int. Ed.* **2018**, *57*, 3943-3948
CuV_2O_6	427	0.1	*ACS Nano* **2019**, *13*, 12081–12089
$Zn_3V_2O_7(OH)_2\cdot 2H_2O$	213	0.05	*Angew. Chem. Int. Ed.* **2017**, *30*, 1705580
$(Mn_{0.13}Zn_{0.03})V_2O_5$	463	0.1	*Small.* **2024**, *21*, 2408596
$V_3O_7\cdot H_2O$	386.7	0.1	*Small.* **2023**, *20*, 2305030
$Ag_xV_3O_7\cdot H_2O$	437	0.1	*ACS Sustainable Chem. Eng.* **2021**, *9*, 3985–3995
Sn–$V_3O_7\cdot H_2O$	408	0.1	*Adv. Mater.* **2024**, *15*, 2404026
$NH_4V_3O_8$	433	0.5	*Adv. Mater.* **2024**, *34*, 2312789
$NH_4V_4O_{10}$	245.8	0.2	*Adv. Mater.* **2024**, *36*, 2400888
	370	0.5	*Angew. Chem. Int. Ed.* **2023**, *62*, e202303529.
	497	0.3	Our Work
Na@$NH_4V_4O_{10}$	610.7	0.5	*Adv. Mater.* **2023**, *33*, 2305700
$NH_4V_4O_{10}$ NFAs/CC	536	0.5	*Small.* **2023**, *19*, 2304462
$V_5O_{12}\cdot 6H_2O$	226.5	10	*Adv. Mater.* **2024**, *35*, 2420686
$Al_{0.34}V_5O_{12}\cdot 2.4H_2O$	407.8	0.2	*Small.* **2022**, *18*, 2204180
V_6O_{13}	360	0.2	*Adv. Mater.* **2019**, *9*, 1900083
$NaCa_{0.6}V_6O_{16}\cdot 3H_2O$	347	0.1	*Adv. Mater.* **2019**, *9*, 1901968
$SbO_2/K_{0.43}V_6O_1$	414.2	1	*Adv. Mater.* **2024**, *34*, 2304798
$CaV_6O_{16}\cdot 2.7H_2O$	379	0.05	*Adv. Mater.* **2025**, *15*, 2404037
$ZnV_6O_{16}\cdot 8H_2O$	365	0.5	*Angew. Chem. Int. Ed.* **2023**, *62*, e202213368
$H_{3.78}V_6O_{13}$	406	0.1	*Adv. Mater.* **2023**, *33*, 2307270

$Na_2V_6O_{16}$	564	0.1	*ACS Appl. Mater. Interfaces* **2024**, *16*, 58587–58597
$Al_{2.65}V_6O_{13}\cdot 2.07H_2O$	571.7	1	*Angew. Chem. Int. Ed.* **2022**, *62*, e202216089
$Ca_{0.67}V_8O_{20}\cdot 3.5H_2O$	466	0.1	*ACS Nano* **2019**, *13*, 14447–14458
$(Na,Mn)V_8O_{20}$	377	0.1	*Adv. Mater.* **2020**, *7*, 2000083
$K_xH_9(PV_{14}O_{42})$	413.3	0.05	*ACS Appl. Energy Mater.* **2024**, *7*, 629–638
MVO+V_2C	389.4	0.5	*Adv. Mater.* **2024**, *34*, 2402071
VNQD/NC	698	0.5	*Adv. Mater.* **2025**, *35*, 2418671
MIL-88B(V)@rGO	479.6	0.05	*Adv. Mater.* **2024**, *34*, 2308319
Ov-CoVO	475.0	0.2	*Adv. Mater.* **2024**, *34*, 2407925
V_2CT_x	358	30	*Adv. Mater.* **2021**, *31*, 2008033
V-EG	553	0.3	*Adv. Mater.* **2024**, *34*, 2311412
alk-V_2CT_x	498.2	0.1	*Adv. Mater.* **2024**, *34*, 2308508
VO-DP	473	0.05	*Adv. Mater.* **2023**, *33*, 2213187
VNTONC	802.5/331.8	0.5/6	*Small.* **2024**, *20*, 2312036
sodium vanadate	302.4	0.05	*Angew. Chem. Int. Ed.* **2024**, *63*, e202317944
nano-vanadium oxide	553	0.1	*Adv. Mater.* **2024**, *36*, 2404796
O_d-VO	400	0.2	*Angew. Chem. Int. Ed.* **2020**, *59*, 2273-2278
KVO	461	0.2	*Nano Lett.* 2021, *21*, 2738–2744
EDA-VO	382.6	0.5	*Adv. Mater.* **2022**, *34*, 2105452
NHVO@CC	517	0.1	*ACS Appl. Mater. Interfaces* **2022**, *14*, 10407–10418
α-MnO_2	238.3	0.2	*Adv. Mater.* **2025**, *35*, 2412370
K-doped α-MnO_2 (KMO)	218.1	0.2	*Ind. Eng. Chem. Res.* **2023**, *62*,16757–16765
isoleucine-α-MnO_2	332.8	0.1	*Adv. Mater.* **2025**, *15*, 2402819
CNT@β-MnO_2	259	3	*Inorg. Chem.* **2024**, *63*, 13100–13109
β-MnO_2	32		
δ-MnO_2-C NA	346.7	0.5	*Small.* **2023**, *19*, 2207517
δ-$MnO_2/Ti_3C_2T_x$	502.2	0.3	*Adv. Mater.* **2024**, *14*, 2304357
Ru−MnO_2	314.4	0.2	*Angew. Chem. Int. Ed.* **2025**, *64*, e202415997
PVP−MnO_2	317.2	0.125	*Angew. Chem. Int. Ed.* **2023**, *62*, e202313163
C@PODA/MnO_2	321	0.1	*Angew. Chem. Int. Ed.* **2022**, *61*, e202212231.
V_O-MnO_2	303	0.3	*ACS Appl. Energy Mater.* 2023, *6*, 6689–6699
Na:MnO_2/GCF	381.8	0.1	*Adv. Mater.* **2020**, *30*, 1907120
$Ce_{in/inter}$-MnO_2	348.8	0.3	*Adv. Mater.* **2024**, *14*, 2304303
CMC-MnO_2	324	0.5	*Adv. Mater.* **2025**, *35*, 2411990
N-coordinated MnO_2	351	0.1	*Angew. Chem. Int. Ed.* **2024**, *63*, e202408414.
MnO_2@MXSC	368	1	*Adv. Mater.* **2024**, *34*, 2401059

MnO_x/polypyrrole nanosheets	408	0.133	*ACS Appl. Mater. Interfaces* **2020**, *12*, 9347–9354
$NaMnO_2$	190	1.0	*Angew. Chem. Int. Ed.* **2024**, *63*, e202409322.
CaMnO-140	485.4	0.1	*Angew. Chem. Int. Ed.* **2024**, *63*, e202412057.
$Ca_{0.10}MnO_2\cdot0.61H_2O$	289	1.5	*Adv. Mater.* **2025**, *35*, 2413993
V_{Mn}-Mn_3O_4@C	280.9	0.1	*Adv. Mater.* **2025**, *35*, 2413711
Mo doping regnant $LaMnO_3$	445	0.5	*Adv. Mater.* **2025**, *35*, 2416652
$ZnMn_2O_4$	239.2	0.1	*Adv. Mater.* **2024**, *34*, 2405680
$ZnMn_2O_4$/Graphene	445	0.1	*Adv. Mater.* **2024**, *34*, 2405551
MoS_2	439.5	0.1	*Angew. Chem. Int. Ed.* **2024**, *63*, e202320075.
MoS_2/graphene	285.4	0.05	*Adv. Mater.* **2021**, *33*, 2007480
Mn-doped MoO_3	366.2	0.2	*Adv. Mater.* **2023**, *19*, 2301351
S-MoO_2	236	0.1	*Adv. Mater.* **2024**, *34*, 2308834
Bi_2Se_3	320	0.1	*Adv. Mater.* **2023**, *35*, 2306269
Bi_2O_2Se	380.3	0.1	*Adv. Mater.* **2024**, *34*, 2406975
P–$NiCo_2O_{4-x}$	361.3	3	*Adv. Mater.* **2018**, *30*, 1802396
CuS@CTMAB	367	0.1	*ACS Appl. Mater. Interfaces* **2022**, *14*, 1126–1137
TAP/$Ti_3C_2T_x$	303	0.04	*Adv. Mater.* **2022**, *34*, 2206812
BBQPH	498.6	0.2	*Nat Commun.* **2023,** *14,* 5235.
BQPH	429	0.1	*J. Am. Chem. Soc.* **2022**, *144*, 10301–10308
6CN-HAT@MXene	413	0.05	*Adv. Mater.* **2024**, *34*, 2316182
Cu-HHTP	260.1	0.1	*Angew. Chem. Int. Ed.* **2023**, *62*, e202218343.
	228	0.05	*Nat Commun.* **2019,** *10,* 4948.
Cu-BTA-H	330	0.2	*ACS Nano* **2023**, *17*, 3077–3087
Cu-TBPQ MOF	371.2	0.05	*Adv. Mater.* **2024**, *34*, 2312636
COF-TMT-BT	283.5	0.1	*Angew. Chem. Int. Ed.* **2023**, *62*, e202216136.
COF-PTO	165	0.3	*Adv. Mater.* **2024**, *36*, 2314050
DQP	413	0.05	*ACS Appl. Energy Mater.* **2021**, *4*, 655–661
DPT	367/303/231	0.1/5/20	*J. Am. Chem. Soc.* 2023, **145**, 25604–25613
DQH	385	0.5	*Adv. Mater.* **2024**, *34*, 2405710
HATNQ	482.5	0.2	*Angew. Chem. Int. Ed.* **2022**, *61*, e202116289.
HATN-PNZ	257	5	*Adv. Mater.* **2023**, *35*, 2207115
PTO	16/142	5/10	*J. Am. Chem. Soc.* **2024**, *146*, 15393–15402.
TNP	338		*Angew. Chem. Int. Ed.* **2024**, *63*, e202401049.
TABQ	303/213	0.1/5	*Nat Commun.* **2021,** *12,* 4424.